Host-Prefecture Top-*k* Concentration at Japan's Kokutai: Descriptive Evidence from 75 Completed Editions (1948–2025) with an Exploratory Sport-Type Comparison (2024–2025)

Running title: Host-prefecture top-*k* concentration across 75 completed Kokutai editions

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Abstract

Background: Host-country advantage in international sport is well established across the Summer and Winter Olympic series (Balmer, Nevill & Williams 2001, 2003; Wilson & Ramchandani 2021), and Balmer, Nevill & Williams (2003) further proposed that home advantage should be concentrated in *subjectively judged* events (boxing, gymnastics) rather than in objectively measured events (track, weightlifting). Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) is an unusually long-running setting for the descriptive counterpart of these questions — 47 prefectures rotating as hosts under centrally administered but periodically revised rules since 1946 — yet the leading Japanese-language predecessor (Funahashi et al. 2016) analyzed only a 9-year window (2003-2011) and did not document the long-run overrepresentation pattern of host prefectures among top-*k* finishers or an objective/subjective decomposition. Using the full editions 3-79 (1948-2025), we ask three descriptive questions: (i) do host prefectures reach top-1, top-3, and top-8 at rates substantially exceeding a *k*/47 chance baseline in which the host role is treated as if drawn uniformly at random from the 47 prefectures (recognizing that this baseline does not identify a causal hosting effect — see Limitations Primary conceptual limitation); (ii) is this host-prefecture top-*k* concentration stable across the 75 completed editions in the 3-79 span (77 editions total, of which 2 were cancelled for COVID-19), or does it vary by era; (iii) in the 2024 and 2025 editions — the only two for which JSPO publicly releases the full 47-prefecture × ~40-sport disaggregation — are per-sport host-versus-non-host score differences larger in subjectively judged sports than in objectively measured ones?

Methods: We assembled a host-prefecture ranking panel spanning the 77 editions from the 3rd (1948) through the 79th (2025) meet, of which 75 were non-cancelled and included in the analysis (n = 150 host-observations = 75 non-cancelled meets × 2 trophies; the emperor's cup for combined male-female standings and the empress's cup for female-only; the 75th (2020) and 76th (2021) editions were cancelled

for COVID-19), parsed from the Nagano Prefecture Sports Association ranked-listings page (independently cross-validated against JSPO official archive PDFs for the 58th-67th editions: 20 of 20 host-year cells matched exactly). Editions 1-2 (1946-1947, no complete official ranking data) and the 75th-76th editions (2020-2021, cancelled for COVID-19) were excluded. The primary analyses were (i) a one-sample proportion test against the chance null of *k*/47 (exact binomial with two-sided Wilson 95% confidence intervals; $k \in \{1, 3, 8\}$), (ii) a permutation test in which host-prefecture identity was re-drawn uniformly at random from the 47 prefectures (n_perm = 10,000, Monte Carlo p-value lower-bound correction (count + 1) / (n_perm + 1) following Phipson & Smyth (2010)), (iii) a three-era global test (conditional Pearson χ^2 exact test via full enumeration of the 2 × 3 contingency tables with fixed marginals, since the shock-era n = 7 violates the χ^2 asymptotic expected-count condition) with pairwise Fisher exact contrasts (Holm-Bonferroni corrected for the 6 per-cup contrasts) partitioning the 75 non-cancelled editions into an early formative era (1948-1977, n = 30), a golden host-victory era (1978-2015, n = 38), and a post-2016 shock era (2016-2025, n = 7), and (iv) a pooled ordered logit of ordinal host rank on era and trophy indicators with cluster-robust standard errors at the host-prefecture level. As a subsidiary analysis, we tested the Balmer et al. (2003) subjective-versus-objective decomposition on a 47-prefecture × 40-sport stacked cross-section of the 2024 Saga and 2025 Shiga editions (n = 6,991 cells; 47 clusters, 2 treated). As a robustness check for the pre-1972 pre-Okinawa-reversion 46-prefecture regime, we recomputed the one-sample tests on the early era under a conservative worst-case 46-state null.

Results: Across all 75 non-cancelled meets, the host prefecture reached the championship in 57 of 75 emperor's cup editions (76.0%; Wilson 95% CI [0.652, 0.842]) and 47 of 75 empress's cup editions (62.7%; [0.514, 0.727]) — respectively 36× and 29× the chance rate of 1/47 = 2.13% (both $p < 10^{-40}$, exact binomial; safe upper cap — the true tail probabilities are far smaller, but a shared cap is reported here in preference to hyper-precise exponents). At the top-3 threshold, the host reached the top 3 in 69 of 75 emperor's cup editions (92.0%) and 64 of 75 empress's cup editions (85.3%), against a chance rate of 3/47 = 6.38%. At the top-8 threshold, both cups reached 93.3% (70/75), against a chance rate of 8/47 = 17.02%. The permutation test corroborated these results with all six threshold-by-cup combinations yielding p = 1/10,001 (the Monte Carlo lower bound). The per-cup three-era global tests (conditional Pearson χ^2 exact tests, full enumeration of the 2 × 3 tables with fixed marginals) revealed a **non-monotonic** time pattern on both cups: for the emperor's-cup top-1 rate, the early era produced 56.7%, the golden era 97.4%, and the shock era 42.9% (**exact $p = 3.6 \times 10^{-5}$**; Pearson χ^2 asymptotic $p \approx 4.8 \times 10^{-5}$ as sensitivity); for the empress's-cup top-1 rate, 30.0%, 92.1%, and 42.9% respectively (**exact $p = 5.4 \times 10^{-8}$**; Pearson χ^2 asymptotic $p \approx 5.2 \times 10^{-7}$). The pairwise Fisher decomposition with Holm-Bonferroni correction (family of six per-cup contrasts) placed all four golden-era contrasts as significant at $\alpha = 0.05$ (three of the four at Holm $p < 0.01$; the fourth, empress's golden-vs-shock, at Holm $p = 2.1 \times 10^{-2}$), while the early-vs-shock contrasts were not detected on either cup (Holm p = 1.00 on both; raw Fisher $p \approx$ 0.68 emperor's / 0.66 empress's) — the early and shock point estimates were similar on the top-1 metric but the shock-era estimates are imprecise (n = 7 editions per cup) and no differences were detected, so these observations should not be read as equivalence results (see Limitations §Third); only the golden era differs sharply from both endpoints on both cups. Both cup-specific ordered logits (n = 75 each) showed

markedly better ranks during the golden era relative to the early-era reference (emperor's cup era_golden = −3.37, SE = 1.20, $p \approx 5 \times 10^{-3}$; empress's cup era_golden = −3.45, SE = 0.88, $p \approx 9 \times 10^{-5}$; negative = better rank), whereas the shock-era estimates diverged between the two cups (emperor's cup era_shock = +0.08, SE = 0.58, p = 0.89; empress's cup era_shock = −1.15, SE = 0.57, p = 0.045); the pooled specification is described in the Results but is not used here because it does not correct for the within-edition correlation of the two cups (see Limitations §Third). The 46-versus-47-state sensitivity check found that the excess above chance differs by less than 0.005 across all early-era threshold-cup cells, so the default 47-state null mildly overstates excess but the impact is marginal. In the 2024-2025 sport-level cross-section, the host × subjective interaction was +16.68 points (cluster-robust SE = 7.72, p = 0.031); the wild-cluster bootstrap (Rademacher weights, restricted null, B = 999) yielded p = 0.155 under the few-treated-clusters correction (only 2 of 47 clusters treated) and should be read as directional rather than decisive. The descriptive host-boost gap ran monotonically from +37.9 points in subjective sports through +26.2 in semi-subjective/team to +17.6 in objective, in the direction Balmer et al. (2003) predicts. Two scale-invariant normalizations produced mixed evidence: the z-score within sport-year-cup cell preserved the positive direction (bootstrap p = 0.158, matching the raw-score bootstrap p = 0.155), whereas the percentile rank within the same cell attenuated to bootstrap p = 0.498.

Conclusions: Across the 75 completed editions in the 3-79 span (1948-2025), host prefectures were substantially overrepresented among top-*k* finishers relative to a *k*/47 uniform-selection chance baseline (top-8 in 93.3% of editions against 17.02% chance; emperor's-cup championship in 76.0% against 2.13% chance). This descriptive concentration was highest during 1978-2015 and lower during the pre-1978 and post-2016 windows, although the precise boundaries and the causal mechanisms of the temporal pattern remain uncertain (see Limitations Primary conceptual limitation and §Third). An exploratory analysis of the 2024 and 2025 editions suggested larger per-sport host-versus-non-host score differences in subjectively classified sports, but the estimate rests on only two host prefectures (Saga in 2024 and Shiga in 2025), is sensitive to the choice of inference method (cluster-robust p = 0.031 vs. wild-cluster bootstrap p = 0.155 under the few-treated-clusters correction), and does not identify judging bias, subject-sport crowd effects, subject-sport practice-environment effects, or any other causal hosting mechanism. Because the confirmatory *k*/47 baseline treats the host role as if drawn uniformly at random from the 47 prefectures — and does not adjust for prefecture-level competitive strength, population, or budget — the descriptive results should be read as a documentation of host-prefecture top-*k* concentration in the observed Kokutai record rather than as evidence for a causal home-advantage effect. Historical sport-level records for editions 1-65 exist in the JSPO's 65-edition compendium but their tabular structure precludes reliable structured parsing at present; extension to a full-panel causal analysis awaiting such data (or a parallel Zitzewitz-style judge-level analysis for the identification of judging bias specifically) is left to future work.

Keywords: Home advantage, host effect, National Sports Festival, Kokutai, one-sample proportion test, permutation inference, non-monotonic time pattern, subjective judging, Balmer 2003, Japan

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Introduction

Does hosting a sporting event genuinely improve the host's performance, and if so, through what mechanism? At the international scale, the answer is unambiguously yes: host countries reliably win more medals than their non-host baseline predicts, an effect documented across a century of Summer Olympic Games (Balmer, Nevill & Williams 2003)[3] and the Winter Olympic series (Balmer, Nevill & Williams 2001)[1], with a broader Olympic/Paralympic replication (Wilson & Ramchandani 2021)[2] over the modern 1988-2018 window. The mechanism, however, is contested. Six candidate pathways are typically invoked — host-country entry quotas, training-venue familiarity, home-crowd support, budget uplift, jet-lag/climate advantage, and, most contentiously, **subjective-judging bias in favor of the host** — but no single mechanism has been uniquely identified in any single dataset.

Balmer, Nevill & Williams (2003) proposed a diagnostic decomposition.[3] If home advantage were driven by facility familiarity, budget, or crowd support, it should appear roughly uniformly across all Olympic sports. If, on the other hand, referee or judge bias were a load-bearing mechanism, home advantage should be concentrated in sports where the outcome depends on human scoring rather than an objective clock or tape measure. Comparing five event families across a century of summer Olympic data, Balmer et al. (2003) found that home advantage is indeed disproportionately concentrated in subjectively judged sports — boxing and gymnastics, in the specific event groups reported in the paper's abstract — with much smaller or absent effects in objectively measured sports such as athletics and weightlifting. This decomposition provides an implicit test of the judging-bias mechanism that does not require access to individual judge scores.

Japan's National Sports Festival (国民体育大会, *Kokutai*; renamed 国民スポーツ大会 in 2024) offers an unusually long-running setting for testing whether these results generalize: a centrally administered rotating-host competition running since 1946, with the host role cycling across all 47 prefectures on a fixed JSPO-managed schedule. Scoring rules and eligibility have been revised episodically over the 80-year period (notably the 2005 *furusato* athlete-affiliation reform noted in the Two host-loss clusters discussion) rather than remaining literally unchanged, so the "same rules since 1946" phrasing that appears in some Kokutai commentary should be read as centrally administered continuity rather than rule stasis. The event is large (approximately 40 sports across a summer meet and a smaller winter meet), long-running (79 numbered editions through 2025; the 1st and 2nd editions (1946 and 1947) have no complete official 47-prefecture ranking data available; the 75th and 76th editions (2020 Kagoshima and 2021 Mie) were cancelled for COVID-19; the analysis window is therefore the 75 completed editions of the 3rd through 79th (n = 77 in the 3-79 span, minus 2 COVID cancellations), with an 80th scheduled for autumn 2026 in Aomori), and centrally administered by the Japan Sport Association (JSPO). Three headline patterns are widely known within Japan. First, host prefectures win at a startling rate: across the full 1948-2025 series (editions 3-79, excluding the never-completely-ranked 1946-1947 first two editions and the 2020-2021 COVID-cancelled meets), the host prefecture reaches the emperor's cup championship in 57 of 75 non-cancelled meets (76.0%). Second, this aggregate figure masks a striking non-monotonic time pattern: from 1978 through 2015 (the "*golden era*" of host wins in the local parlance), the host prefecture took the emperor's cup championship in **37 of 38 editions (97.4%)** and the empress's cup in **35 of 38 (92.1%)**, whereas in the post-2016 shock era the host championship rate has collapsed to **3 of 7 (42.9%)** on both cups across the 7 non-cancelled meets in 2016-2025 (the 2020 Kagoshima and 2021 Mie editions were cancelled for COVID-19). The pre-1978 early-era host championship rate was **17 of 30 (56.7%) on the emperor's cup and 9 of 30 (30.0%) on the empress's cup**, so the shock era's rate is close to the early-era emperor's-cup rate but higher than the early-era empress's-cup rate on the point-estimate level; per-cup pairwise Fisher exact tests do not detect a difference between early and shock on either cup (emperor's cup $p \approx 0.68$, empress's cup $p \approx 0.66$), which reflects the shock era's small sample (7 non-cancelled editions per cup) rather than positive evidence of equivalence (see Limitations §Third). Both early and shock differ sharply from the golden era on both cups. Third, the post-2016 collapse has generated both journalistic speculation and one scholarly critique (Suetsugu, 2024, 2025)[4,5] that host outcomes reflect a "host-victory-first mindset" (a phrase corresponding to the "開催地優勝至上主義" framing that appears in the title of Suetsugu 2025[5]). (For the historical record, the popular phrase "*Nara hantei*" — a slur for perceived hometown scoring bias — refers to a 2018 boxing scandal centered on Yamane Akira, then president of the Japan Amateur Boxing Federation, as documented in Japanese national news coverage of the federation's July 2018 self-dissolution and subsequent JOC/JSPO governance reforms; the phrase does not refer to the Kokutai kendo competition as is sometimes claimed.)

The most substantial quantitative predecessor is Funahashi, Hibino, Ishiguro & Mano (2016).[6] They assembled a balanced 47-prefecture × 9-year panel (n = 423) for 2003-2011, regressed the male-female combined competition score on a full set of host-year event-time dummies (t-7 through t+7), and estimated a host-year bonus of +1,674.65 total-score points on top of prefecture and year fixed effects ($R^2 = 0.86$). Funahashi (2016)'s Table 4 lists the six candidate mechanisms invoked above and cites Balmer et al. (2001)[1] — Balmer's Winter Olympics paper — as their reference for the subjective-judging pathway. But their analysis is bounded in three separate ways relative to what a full assessment of the Kokutai host effect requires. First, they **did not extend the analysis window beyond 2011**, and consequently could not test whether the host bonus is stable across eras — the post-2016 collapse noted above lies entirely outside their sample. Second, they **did not test the reality of the host bonus against a formal chance baseline** (such as the k/47 uniform-selection null against which host performance in the top-k is naturally benchmarked); their fixed-effects regression documents that host years score higher than non-host years for the same prefecture but does not directly test whether the host prefecture's top-k finishing rate exceeds what would occur under random assignment of the host role. Third, they **did not cite Balmer et al. (2003) or include either a sport-type indicator or a host × sport-type interaction** in any of their specifications, and the subjective-versus-objective decomposition that Balmer et al. (2003) formalized is therefore absent from the Japanese panel literature. The 75-completed-edition reality test (span editions 3-79), the era-heterogeneity analysis, and the Balmer et al. (2003) subjective-versus-objective decomposition are the three specific gaps this paper addresses. Chiba's earlier qualitative treatment (1987)[7] likewise enumerates five host-victory mechanisms — the full-entry system, alleged combination-drawing bias, transferred-athlete contributions, intra-prefecture athlete development, and prefecture-wide civic support — and explicitly does not mention refereeing or judging bias in any of the five.

Institutionally, the Japanese Kokutai stands in mild contrast to the Korean equivalent. The Korean Sport & Olympic Committee's national comprehensive sports competition regulations explicitly award a **20% bonus** to the host city's overall score, codified from 2010 onward (previously 10% from 2001).[8] Japan has no such explicit host bonus in its scoring rules — the JSPO regulations we examined contain no analogous provision — which makes the Japanese host advantage, if present, a purely emergent phenomenon rather than an institutional artifact. This makes the reality test especially informative in the Japanese case: a host bonus that appears at extreme magnitudes in the absence of an institutional multiplier is more strongly suggestive of the constellation of behavioral mechanisms (host-quota selection, facility familiarity, crowd support, and — potentially — subjective-judging bias) that Balmer et al. (2003) and the subsequent literature have proposed.

We accordingly ask three questions. First (**reality test**), does the host prefecture reach top-*k* finishing positions at rates significantly exceeding the pure-chance baseline of *k*/47 across the 75 completed editions in the 3-79 span (77 editions total, of which the 75th (2020 Kagoshima) and 76th (2021 Mie) were cancelled for COVID-19), and does this reality survive across the top-1 (championship), top-3, and top-8 thresholds so that the finding is not fragile to a single threshold choice? Second (**era heterogeneity**), is the host bonus stable across these 75 completed editions, or does it vary systematically by era — and if the latter, is the variation monotonic (a smooth trend from 1948 through 2025) or non-monotonic (with the pre-1978 early era and the post-2016 shock era resembling each other more than either resembles the 1978-2015 golden era)? Third (**subjective-versus-objective decomposition**), following Balmer et al. (2003), is the host bonus larger in subjectively judged sports (boxing, judo, gymnastics, kendo, karate, …) than in objectively measured ones (track, swimming, cycling, weightlifting, …) in the 2024 Saga and 2025 Shiga editions — the only two editions for which JSPO publicly releases the full 47-prefecture × ~40-sport disaggregation? Our answers, in brief, are: (i) **yes, descriptively and consistently across three thresholds** — for the emperor's cup the host reaches the championship in 76.0% of the 75 non-cancelled editions versus the chance rate of 2.13% (both cups $p < 10^{-40}$ by one-sample exact binomial; the top-3 rate is 92.0% versus 6.38%, and the top-8 rate is 93.3% versus 17.02%, so the concentration pattern is stable across all three thresholds). We stress from the outset that this baseline treats the host role as if drawn uniformly at random from the 47 prefectures; it therefore documents a descriptive overrepresentation of host prefectures among top-*k* finishers rather than identifying a causal hosting effect that would require holding fixed prefecture-level competitive strength, population, budget, and pre-hosting athlete-development trajectories (see Limitations Primary conceptual limitation). (ii) **non-monotonic across author-defined era boundaries** on the top-1 metric — the early formative era (emperor's cup 56.7%, empress's cup 30.0%; 1948-1977) and the post-2016 shock era (both cups 42.9%; 2016-2025) exhibit point estimates well below the golden-era peak on both cups, and per-cup pairwise Fisher exact tests do not detect a difference between early and shock on either cup (emperor's Fisher $p \approx 0.68$, empress's Fisher $p \approx 0.66$) — a non-detection that reflects the shock era's small sample ($n = 7$ non-cancelled editions per cup, so the width of the shock-era 95% CI is substantial and a non-rejection cannot be read as an equivalence result — see Limitations §Third) — whereas the 1978-2015 golden era (emperor's 97.4%, empress's 92.1%) differs sharply from both endpoints on both cups (per-cup conditional exact tests: emperor's cup exact $p = 3.6 \times 10^{-5}$; empress's cup exact $p = 5.4 \times 10^{-8}$). The era boundaries themselves (1978 as the golden-era onset and 2016 as the shock-era onset) were selected with knowledge of the outcome data and are not pre-registered (see Limitations §Third and Supplementary Materials §S4 boundary sensitivity). (iii) **directionally consistent with Balmer et al. (2003) as an exploratory two-year comparison** — the descriptive per-sport host-boost gap runs monotonically from +37.9 points in subjective sports through +26.2 in semi-subjective/team to +17.6 in objective (Table 7); the regression host × subjective interaction is +16.68 points (cluster-robust $p = 0.031$), but under the appropriate few-treated-clusters wild-cluster bootstrap correction (only 2 of 47 prefecture clusters treated: Saga in 2024 and Shiga in 2025) the interaction does not attain conventional significance (bootstrap $p = 0.155$). This exploratory finding should be read as directional evidence in the Balmer et al. (2003) predicted direction rather than as identification of judging bias, subject-sport crowd effects, or any other specific causal mechanism, and rests on only two host prefectures.

Methods

Data sources

The primary analysis dataset is a host-prefecture ranking panel of all 75 non-cancelled editions in the span from the 3rd (1948) through the 79th (2025) meet (77 editions in total in this span, of which 2 were cancelled), yielding $n = 150$ host-observations (75 non-cancelled meets × 2 trophies: the emperor's cup (天皇杯) for combined male-female standings and the empress's cup (皇后杯) for female-only standings). The 1st (1946) and 2nd (1947) editions were excluded because no complete official ranking data are available (see Limitations); the 75th (2020 Kagoshima) and 76th (2021 Mie) editions were excluded because they were cancelled for COVID-19. Host-prefecture ranks were parsed from the Nagano Prefecture Sports Association ranked-listings page (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html), which reports the top 8 finishers per edition per trophy across the full editions 3-79 window. To validate the primary source, we independently cross-checked host-prefecture ranks against JSPO official archive PDFs (https://www.japan-sports.or.jp/kokutai/tabid183.html) for the 58th-67th editions (2003-2012) — the range for which the JSPO machine-readable PDFs are reliably parsable with `pdfplumber` (Python 3.14) — and confirmed that **20 of 20 host-year cells (10 editions × 2 trophies) matched exactly** between the two sources. For editions 68-77 (2013-2022), the corresponding JSPO PDFs were either 404 responses or image-only files from which text could not be extracted; the Nagano source provides host-prefecture ranks for these editions and is used as the sole source in that range. For editions 78-79 (2024 Saga and 2025 Shiga), JSPO also published Excel disaggregations of the full 47-prefecture × ~40-sport breakdown (`78_score_data.xls`, `78_score_all_data.xls`, `79_score_all_data.xls`, `79_score_deta.xls`), parsed with `python-calamine` (the Rust-based `xlrd` replacement, required because both files use a formula function ID that `xlrd` misidentifies); these Excel files are used only for the subsidiary sport-level cross-section (Study design item Subsidiary, below), not for the primary host-rank panel.

Because the 3rd-79th editions include three multi-prefecture co-hosted meets (the 7th in 1952 Fukushima/Miyagi/Yamagata, the 8th in 1953 Ehime/Kagawa/Tokushima/Kochi, and the 48th in 1993 Kagawa/Tokushima), the primary host-rank panel normalizes each co-hosted edition to its single administrative host prefecture (defined by the JSPO official record: Fukushima for the 7th, Ehime for the 8th, and Kagawa for the 48th). For the permutation test only (Study design item 2, below), the co-hosted definition is relaxed to "any co-hosting prefecture in the top *k*", to match the pooled 1-of-47 chance null against which co-hosting inflates the probability of at least one co-host reaching the top. Both definitions yielded the same observed top-1 count on the current sample (57 emperor's-cup and 47 empress's-cup out of 75 non-cancelled meets), so the primary results do not depend on which definition is used; the definitional difference is documented in `src/analysis_main_v3.py:permutation_test`.

For the subsidiary 2024-2025 sport-level cross-section, sports were classified as **objective** (n = 17: track and field, swimming, cycling, rowing, canoe, weightlifting, sailing, ski, skating, archery, kyudo, rifle shooting, clay-target shooting, golf, bowling, triathlon, sport climbing; clay-target shooting was appended in the 2024 edition and removed in 2025, so the effective objective count is 17 in 2024 and 16 in 2025), **subjective** (n = 11: kendo, judo, gymnastics, karate, jukendo, naginata, boxing, fencing, wrestling, sumo, equestrian), or **semi-subjective / team** (n = 13: team ball sports with referee-mediated but primarily objective outcomes: soccer, basketball, volleyball, handball, hockey, rugby, softball, baseball, badminton, table tennis, soft tennis, tennis, ice hockey), following Balmer, Nevill & Williams (2003).[3] The classification is implemented in `src/sport_classifier.py` and unit-tested against the Balmer et al. (2003) decision rules.

Study design

The primary analysis is organized around four specifications applied to the 150 host-observations (75 non-cancelled meets × 2 trophies), complemented by one subsidiary specification on the 2024-2025 sport-level cross-section and one sensitivity check on the pre-1972 46-state regime.

1. One-sample proportion test (primary reality test). For each combination of threshold $k \in \{1, 3, 8\}$ and cup ∈ {emperor's, empress's}, we test the null hypothesis that the host prefecture's top-k rate equals $k/47$ (the chance rate under uniform random selection of the host role from the 47 prefectures) against the one-sided alternative that the true rate exceeds $k/47$. The pooled-across-cups count (n = 150) is reported in the accompanying text as a descriptive summary only and is not used as a primary formal test because the emperor's cup and empress's cup within the same edition share the host prefecture, athletes, and venues and are not independent observations (see Limitations §Third). Test statistic: the exact binomial tail probability (`scipy.stats.binomtest(..., alternative='greater')`). Confidence intervals: two-sided Wilson 95% CI (`scipy.stats.binomtest(...).proportion_ci(0.95, method='wilson')`), reported directly rather than as one-sided CIs from the same test to avoid the trivial-upper-bound-1.0 artifact that a one-sided binomial CI would produce. Sample sizes per cup: 75 non-cancelled meets in total; 30 in the early era (1948-1977), 38 in the golden era (1978-2015), 7 in the shock era (2016-2025). The specification is deliberately covariate-free: the confirmatory reality test asks only whether the observed host top-*k* rate exceeds the *k*/47 chance baseline, not whether that exceedance is attributable to any particular set of confounders. Covariate adjustment addresses a different causal question and was outside the scope of this descriptive benchmark.

2. Permutation test (statistical robustness). For each threshold-cup combination, host-prefecture identity is re-drawn uniformly at random from the 47 prefectures independently for each edition (n_perm = 10,000; the null draws are i.i.d. per edition, so the permutation null is a bootstrap-of-labels rather than a fixed-margin permutation of the same 75 hosts). The Monte Carlo p-value is computed with the Phipson & Smyth (2010) lower-bound correction: p_perm = (count + 1) / (n_perm + 1), so a value of 1/10001 ≈ 10^{-4} is the minimum observable p-value and should be read as "no null draw among 10,000 produced an outcome at least as extreme as observed" rather than as a literal 10^{-4} point estimate of the true tail probability. The three co-hosted meets (7th 1952 Fukushima/Miyagi/Yamagata, 8th 1953 Ehime/Kagawa/Tokushima/Kochi, 48th 1993 Kagawa/Tokushima) enter the permutation null with a single at-random draw per edition to preserve the 1-of-47 chance baseline, matching the one-sample test's normalization to a single administrative primary host. On the current sample, the primary-host-only and co-host-inclusive definitions yield the same observed top-*k* count on all three thresholds (top-1, top-3, top-8) and on both cups, so the two definitions produce identical Table 3 observed rows; the definitional difference is documented in Data sources above and in `src/analysis_main_v3.py::permutation_test`.

3. Three-era global test with pairwise Fisher exact contrasts (era heterogeneity). The 75 non-cancelled editions are partitioned into three author-defined eras: **early** (1948-1977, kai 3-32, 30 meets), **golden** (1978-2015, kai 33-70, 38 meets), and **shock** (2016-2025, kai 71-79 excluding cancelled 75-76, 7 meets). The boundaries were selected with knowledge of the observed outcome data (1978 corresponds to the widely-cited onset of the "golden era" of host wins in the Japanese Kokutai literature; 2016 to the first host loss in more than a decade). Because the shock era has only n = 7 editions per cup, the expected cell counts under the joint-independence null fall below 5 in several 2 × 3 cells (e.g., empress's cup top-1 shock cell: expected non-success ≈ 2.61, expected success ≈ 4.39), so the standard Pearson χ^2 asymptotic approximation is not reliable. We therefore report a **conditional Pearson χ^2 exact test** obtained by fully enumerating all 2 × 3 contingency tables with fixed row and column marginals (100-600 tables per cell) and summing the multivariate hypergeometric probabilities of tables whose Pearson χ^2 statistic equals or exceeds the observed statistic (implemented in `src/analysis_main_v3.py::_conditional_exact_p_chi2`); this is the **primary global test** for Table 4. As a secondary numerical check we also report a Monte Carlo permutation p on the same statistic (n_perm = 10,000; Phipson & Smyth (2010) lower-bound correction), which matches the enumerated exact p to within 2×10^{-2} across all six primary rows. The Pearson χ^2 asymptotic p (`scipy.stats.chi2_contingency`; the Yates continuity correction is inapplicable to 3 × 2 tables and is not invoked by scipy in this case) is retained as a sensitivity column. The pairwise Fisher exact tests (`scipy.stats.fisher_exact`) are further Holm-Bonferroni corrected across the six per-cup contrasts (2 cups × 3 era-pairs) so that the non-monotonic vs. monotonic character of any era heterogeneity can be read directly from the multiple-comparisons-adjusted pairwise results. A boundary-sensitivity analysis (golden-start ∈ {1975..1980} × shock-start ∈ {2014..2018} × 2 cups = 60 combinations) is reported in Supplementary Materials §S4 (e.g., a monotonic time trend would produce three ordered pairwise contrasts, whereas a non-monotonic pattern would produce two significant pairwise contrasts flanking one non-significant contrast between the two endpoint eras). The three-era partition is historically motivated but was selected with knowledge of the observed outcome pattern rather than pre-registered from historical criteria alone (the 1978 boundary corresponds to the widely-cited onset of the "golden era" of host wins in the Japanese Kokutai literature, and the 2016 boundary corresponds to the first host loss in more than a decade — 2016 Iwate); the era analysis is therefore descriptive and exploratory rather than pre-registered or confirmatory, no formal preregistration record (OSF, AsPredicted, or equivalent timestamped protocol) accompanies the partition, and the partition itself is a researcher degree of freedom whose consequences are discussed in Limitations §Third. The ordered-logit specification below uses the same three-era categorical variable and therefore inherits the same boundary-selection dependency.

4. Pooled ordered logit (era heterogeneity, regression form). As a regression-based cross-check on the era-heterogeneity result, we fit a pooled ordered logit of ordinal host rank on era and trophy indicators (with fixed effects (FE) for era and cup as detailed below): `rank_ordinal ~ C(era, Treatment('early')) + C(cup, Treatment('kougou'))`, where `rank_ordinal` is the host's finishing rank censored at 9 (a placeholder for "outside top 8"), `era` is a three-level categorical with `early` as the reference, and `cup` is a two-level categorical with the empress's cup as the reference. The specification is fitted in three modes: (a) pooled across both cups (n = 150, cup as a covariate); (b) emperor's cup only (n = 75, cup dropped); (c) empress's cup only (n = 75, cup dropped). Standard errors are clustered at the host prefecture code (`host_pref_code`; the 46 distinct host prefectures represented in the 75-meet series (46 of the 47 Japanese prefectures have hosted at least one Kokutai in the 3rd-79th editions)) to account for repeated hosting by the same prefecture. The ordered logit is fitted with `statsmodels.miscmodels.ordinal_model.OrderedModel`, and a graceful fallback to Fisher-information (unclustered) SEs is implemented for the rare case in which the cluster meat matrix is singular (not triggered on the current sample; the graceful-fallback code path is not currently unit-tested — see Limitations).

Subsidiary. 2024-2025 sport-level cross-section (Balmer et al. (2003) subjective-versus-objective decomposition). The Balmer et al. (2003)-style host × sport-type interaction is tested on a stacked cross-section of **n = 6,991 prefecture-sport-year cells** from the 2024 Saga and 2025 Shiga editions (47 prefectures × ~40 sports × 2 years × 2 trophies; the theoretical full cartesian product would be 47 × [(40 tennou + 35 kougou)_2024 + (40 tennou + 36 kougou)_2025] = 47 × 151 = 7,097 cells, and the regression frame retains 6,991 after dropping 106 unbalanced cells for prefectures that did not participate in specific sports and are absent from the JSPO overall-standings publication for that sport-year, a 1.5% shortfall). The specification is `score_isc = α + β_H · host_isc + β_HS · (host × subj)_isc + η_i + κ_c + ν_t + ξ_p + ϵ_isc` with prefecture (`η_i`), sport (`κ_c`), year (`ν_t`), and trophy (`ξ_p`) fixed effects; prefecture-clustered robust standard errors (47 clusters, Liang-Zeger sandwich). The sport-fixed-attribute main effect `β_S · subj_c` is not included because `subj_c` is a sport-level constant (`subj_c` takes the same value for all prefecture-year-trophy observations of sport `c`) and is therefore an exact linear combination of the sport fixed effects `κ_c`; only the host × subjective interaction is identified. The 13 semi-subjective/team sports are grouped with the 17 objective sports as the reference category (`subj_c = 0`), so β_H is the host bonus for the combined objective + semi-subjective/team reference group. See Limitations §Fourth for the identifying-variance implications of this binary coding on the 2024-2025 sample. Because only 2 of 47 clusters are treated (Saga in 2024, Shiga in 2025), the cluster-robust p-value is complemented by a wild-cluster bootstrap (Rademacher weights, restricted null, B = 999, fixed seed for reproducibility) as the appropriate few-treated-clusters correction (Cameron, Gelbach & Miller 2008; MacKinnon & Webb 2017). We treat this subsidiary specification as directionally informative rather than as a formal confirmatory test given the two-treated-clusters structure. A log-outcome specification (`log(1 + score)`) is reported as a cluster-robust robustness check on functional form.

Sensitivity check. Pre-1972 46-state regime. Okinawa did not rejoin Japan until 1972, so editions 3-26 (1948-1971) had only 46 participating prefectures rather than 47. To bound the effect of this small change in the chance baseline, we re-compute all early-era one-sample proportion tests under a 46-state null (`null_rate = k/46` in place of `k/47`), applied **uniformly across the full 30-edition early era** (a conservative worst-case bound rather than an exact recompute restricted to the 24 pre-1972 editions; the additional 6 editions from 1972-1977 had 47 participating prefectures, so applying the 46-state null to them slightly under-states the chance baseline for those 6 specific years but produces a strict upper envelope for the 46-vs-47 effect-size difference across the full early era). Because `null_rate = k/n_states` is a decreasing function of `n_states`, the default 47-state null yields a slightly larger `excess = obs_rate − null_rate` than the 46-state null; a marginal excess-difference between the two nulls therefore indicates that the default 47-state analysis *mildly overstates* excess (not understates), and is the direction relevant to a Limitations claim of the form "the default 47-state analysis is a slight upper bound on the true early-era excess".

Statistical software

All analyses were performed with Python 3.14 using pandas 3.0.5, numpy 2.5.1, statsmodels 0.14.6, patsy 1.0.2, scipy 1.18.0, python-calamine 0.8.2, pdfplumber 0.11.10, matplotlib 3.11.1, pyarrow 25.0.0, pytest 9.1.1, and (for the PDF build pipeline) weasyprint 69.0 and markdown 3.10.2. The primary one-sample proportion tests use `scipy.stats.binomtest` with `alternative='greater'` for one-sided p-values and Wilson two-sided 95% CIs; the permutation tests use `numpy.random.default_rng` (fixed seed for reproducibility) with the Phipson & Smyth (2010) Monte Carlo lower-bound correction; the three-era global test uses a conditional Pearson χ^2 exact test via full enumeration (implemented in `src/analysis_main_v3.py::_conditional_exact_p_chi2`; 2 × 3 tables enumerate to 100-600 candidate tables under fixed marginals) as the primary p-value — because the shock era's n = 7 violates the Pearson χ^2 asymptotic expected-count condition — with a Monte Carlo permutation p on the same statistic (`::_mc_permutation_chi2`, n_perm = 10,000) reported as a numerical cross-check, `scipy.stats.chi2_contingency` (Yates continuity correction inapplicable to 3 × 2 tables) reported as an asymptotic sensitivity, and `scipy.stats.fisher_exact` with Holm-Bonferroni correction across the six per-cup contrasts (`statsmodels.stats.multitest.multipletests`) for pairwise 2 × 2 tests; the pooled ordered logit uses `statsmodels.miscmodels.ordinal_model.OrderedModel` with cluster-robust standard errors at the host prefecture code (46 unique clusters — 46 of the 47 Japanese prefectures have hosted at least one Kokutai in the 3rd-79th editions). For the subsidiary 2024-2025 cross-section, prefecture-cluster-robust standard errors (47 clusters, Liang-Zeger sandwich) are reported alongside a wild-cluster bootstrap (Rademacher weights, restricted null, B = 999) as the few-treated-clusters correction. The one-sample proportion tests are one-sided at $\alpha = 0.05$ because the alternative hypothesis (host exceeds chance) is directional by construction; all other statistical tests are two-sided at $\alpha = 0.05$ unless otherwise noted. Unit tests (n = 350 total, all passing; runtime ≈ 190 s on Python 3.14) verify the numerical integrity of each analysis module against small deterministic fixtures.

Multiple comparisons. The confirmatory test of this paper is the one-sample proportion test of the host top-k rate against the k/47 chance null (Question (i), reality test), which is inherently a family of six tests across the three thresholds $k \in \{1, 3, 8\}$ and two cups (emperor's and empress's). All six test statistics reject the null at Monte Carlo minimum p = 1/10001 under the permutation robustness check, so no p-value adjustment is required for the reality test to survive any conventional multiple-comparisons correction (under a Bonferroni correction with $\alpha_{family} = 0.05$ across all six one-sample tests, the required per-test threshold is $\alpha_{single} \approx 0.0083$, and all six observed one-sample p-values are far below this threshold). The era heterogeneity test (Question (ii)) is a single joint 3-era global test per threshold-cup pair, complemented by three pairwise Fisher exact contrasts, and is treated as an exploratory global comparison per cup/threshold rather than as a confirmatory family (the era boundaries themselves are author-defined rather than pre-registered — see Limitations §Third and Supplementary §S4 boundary sensitivity); the paper's headline era-heterogeneity claim (non-monotonic pattern with the early and shock eras similar in point estimate but the shock era's small sample precluding an equivalence result, and only the golden era differing sharply from both) rests on the two per-cup top-1 global tests (conditional Pearson χ^2 exact $p = 3.6 \times 10^{-5}$ for emperor's cup and 5.4×10^{-8} for empress's cup, with Pearson χ^2 asymptotic sensitivity $p \approx 4.8 \times 10^{-5}$ and 5.2×10^{-7} respectively) rather than on the both-cups pooled statistic (removed as non-independent; see Limitations §Third) and does not depend on p-value adjustment across thresholds. The subsidiary Balmer decomposition (Question (iii)) is treated as an *exploratory* interaction test on the 2024-2025 cross-section (consistent with the exploratory framing of the paper's title and Abstract regarding the sport-type comparison); because it fails to reject under the appropriate few-treated-clusters wild-cluster bootstrap (p = 0.155), no multiple-comparisons adjustment for the descriptive within-category boost figures is required.

Ethical considerations

This study analyzes publicly available Kokutai overall-standings data released by the Japan Sport Association and prefectural sport associations. No individual-level, health, or personally identifying data are used, and no human subjects were recruited or contacted. Because the analysis is restricted to already-public aggregate competition results without any personally identifiable information, it does not meet the standard threshold for human-subjects research under either domestic (e.g., MEXT / MHLW — the Ministry of Education, Culture, Sports, Science and Technology and the Ministry of Health, Labour and Welfare — ethical guidelines for research using publicly available secondary data) or international (e.g., 45 CFR §46.102 for non-human-subjects data) frameworks; accordingly, no institutional review board approval or waiver was sought or required.

Results

Descriptive statistics: three-era decomposition

The host-rank panel contains 150 prefecture-cup observations (75 non-cancelled editions × emperor's cup and empress's cup; the 1st, 2nd, 75th, and 76th editions are excluded from the analysis window for the reasons stated in Methods). Table 1 summarizes host top-1 / top-3 / top-8 rates by cup and by three eras (early 1948-1977, 30 editions/cup; golden 1978-2015, 38 editions/cup; shock 2016-2025, 7 editions/cup).

Two features stand out immediately. First, in the golden era the host prefecture won the emperor's cup in 37 of 38 editions (97.4%) and the empress's cup in 35 of 38 (92.1%); in the same era the host reached the top 3 in 37 of 38 editions on both cups (97.4%) and the top 8 in 37 of 38 (97.4%). Second, in the early era and in the shock era the top-1 host rate falls to comparable magnitudes on both cups (early 43.3% pooled across cups; shock 42.9%), while the top-3 and top-8 rates remain high (early top-3 = 75.0%, top-8 = 86.7%; shock top-3 = 100.0%, top-8 = 100.0%). The three-era time pattern is therefore non-monotonic on the top-1 metric — the host top-1 rate rises sharply from early to golden, then collapses back to an early-era-comparable level in the shock era — while remaining monotonically increasing (or roughly flat at the ceiling) on the top-3 and top-8 metrics. Figure 1 displays the individual host-rank observations across the full 1948-2025 window with era colour coding; Figure 2 displays the era × cup × threshold cell means with 95% Wilson two-sided confidence intervals.

Reality test: host top-k prevalence versus chance baseline

Table 2 reports the one-sample proportion test comparing observed host top-k rates against a chance-only null rate of k / 47 (top-1: 2.13%; top-3: 6.38%; top-8: 17.02%), per-cup for the six primary confirmatory rows (2 cups × 3 thresholds). For the **emperor's cup**, the observed host top-1 rate is 0.760 (57 of 75; Wilson 95% CI [0.652, 0.842]; excess 0.739), top-3 is 0.920 (69 of 75; CI [0.836, 0.963]; excess 0.856), and top-8 is 0.933 (70 of 75; CI [0.853, 0.971]; excess 0.763); the exact-binomial one-sided upper-tail p is below any conventional threshold on all three, so we report the shared safe cap **$p < 10^{-40}$** (the true tail probabilities are all in the range 10^{-47} to 10^{-79} and the safe cap is chosen to cover the least extreme of the six rows — the two top-8 rows at ≈ 1.02×10^{-47} — while avoiding dependence on rounded numeric output on the more extreme top-1 / top-3 rows). For the **empress's cup**, the observed host top-1 rate is 0.627 (47 of 75; CI [0.514, 0.727]; excess 0.605), top-3 is 0.853 (64 of 75; CI [0.756, 0.916]; excess 0.789), and top-8 is 0.933 (70 of 75; CI [0.853, 0.971]; excess 0.763); the same safe cap $p < 10^{-40}$ applies. Per-cup reporting is the primary specification because the emperor's cup and empress's cup within the same edition share the host prefecture, athletes, and venues and are not independent observations (see Limitations §Third); a naive both-cups pooled n = 150 exact binomial would understate p-values by treating dependent observations as independent, so we do not report the pooled statistic as the primary. For a summary descriptive view only, the both-cups combined counts are 104 of 150 (top-1), 133 of 150 (top-3), and 140 of 150 (top-8), consistent with the per-cup rates above.

Restricted by era and cup, the pattern reappears with era-specific magnitudes across all six per-cup × per-threshold cells (Table 2 detail dumped in `results/analysis_main_v3.txt` [2]). For orientation on the top-1 metric: the emperor's-cup early-era rate is 0.567 (17 of 30), golden 0.974 (37 of 38), shock 0.429 (3 of 7); the empress's-cup rates are 0.300, 0.921, and 0.429 respectively. All era-restricted per-cup p-values (18 additional cells beyond the six primary rows) sit below the Bonferroni-corrected per-test threshold $\alpha\_single = 0.05 / (6 + 18) \approx 0.0021$ under the joint family. The excess above chance is present in every era × cup × threshold cell of the extended dump and is not the artifact of a single dominant era.

As a sensitivity check for the 1972 Okinawa-reversion period (when Japan comprised 46 prefectures rather than 47), we recomputed the early-era comparisons under a 46-state null (null rate = k / 46 rather than k / 47). Because null_rate = k / n_states is inversely proportional to n_states, the default 47-state null slightly overstates excess above chance for the pre-1972 window; under the 46-state null the observed excess differs from the 47-state baseline by less than 0.005 across every threshold-cup cell (top-1 tennou early: 0.5454 vs. 0.5449; top-3: 0.7695 vs. 0.7681; top-8: 0.6965 vs. 0.6928). The impact is therefore marginal and does not affect any qualitative reading of Table 2 (see Limitations and Supplement §S3 for the data appendix).

Permutation inference

As a numerical implementation check on the $k/47$ uniform-selection benchmark used by the one-sample tests, we conducted a permutation simulation that repeatedly reassigns the host label uniformly at random over the 47 prefectures within each edition and recomputes the top-k host rate under the shuffled labels (10,000 permutations; Table 3). The permutation-null top-k rates concentrated tightly around the theoretical chance rate (e.g., top-1 emperor's cup: null mean 0.021, null SD 0.017, null 95th percentile 0.053), and the observed host top-k rates exceeded the 95th percentile of the permutation null by wide margins on every threshold × cup cell (top-1 emperor's cup: observed 0.760 vs. null 95th 0.053; top-3 empress's cup: observed 0.853 vs. null 95th 0.107). The empirical count of permutations reaching or exceeding the observed rate was zero on every cell; with the Phipson & Smyth (2010)[17] Monte Carlo lower-bound correction applied uniformly ((count + 1) / (n_perm + 1)), we report **p_perm = 1 / 10,001 ≈ 9.999×10^{-5}** for all six threshold × cup cells. This simulation reproduces the same uniform-random-host null the one-sample proportion tests analytically evaluate and confirms the numerical implementation of the *k*/47 benchmark; it does not preserve prefecture-specific competitive strength or the observed non-random hosting schedule, and it is therefore not an independent robustness check against a stable-heterogeneous-prefecture null (see Limitations Primary conceptual limitation).

Era heterogeneity: three-era χ^2 and pairwise Fisher

Table 4 reports the three-era global test with pairwise Fisher-exact follow-ups (Holm-Bonferroni corrected across the six per-cup contrasts), per-cup for the primary formal tests. Because the shock era has only n = 7 editions per cup and the expected cell counts under the joint independence null fall below 5 in several cells (e.g., empress's cup top-1 shock cell: expected non-success ≈ 2.61, expected success ≈ 4.39), the standard Pearson χ^2 asymptotic approximation is not reliable for these tables; we therefore report a **conditional Pearson χ^2 exact test** obtained by fully enumerating all 2 × 3 contingency tables with fixed row and column marginals (100-600 candidate tables per cell) and summing the multivariate hypergeometric probabilities of tables whose Pearson χ^2 equals or exceeds observed as the **primary global test**. A Monte Carlo permutation p on the same statistic (n_perm = 10,000) is reported as a numerical cross-check, and the Pearson χ^2 asymptotic p is retained as a sensitivity column. On the top-1 metric, the **emperor's-cup** three-era comparison rejects the joint null at **exact $p = 3.6 \times 10^{-5}$** (Pearson χ^2 asymptotic $p \approx 4.8 \times 10^{-5}$ as sensitivity), and the **empress's-cup** comparison rejects at **exact $p = 5.4 \times 10^{-8}$** (Pearson χ^2 asymptotic $p \approx 5.2 \times 10^{-7}$). The pairwise Fisher decomposition (2 × 2 exact tests that do not share the shock-era expected-count problem of the 2 × 3 Pearson χ^2), Holm-Bonferroni corrected across the six per-cup contrasts (2 cups × 3 era-pairs), shows for the emperor's cup: early-vs-golden Holm $p = 1.9 \times 10^{-4}$ and golden-vs-shock Holm $p = 4.4 \times 10^{-3}$ (early 56.7% ≪ golden 97.4%; golden 97.4% ≫ shock 42.9%), and for the empress's cup: early-vs-golden Holm $p = 5.4 \times 10^{-7}$ and golden-vs-shock Holm $p = 2.1 \times 10^{-2}$ (early 30.0% ≪ golden 92.1%; golden 92.1% ≫ shock 42.9%). All four golden-era contrasts survive the Holm correction at conventional $\alpha = 0.05$. The **early-vs-shock difference is not detected on either cup** (raw Fisher p ≈ 0.68 emperor's / 0.66 empress's; Holm-corrected p = 1.00 for both) — the pairwise non-rejection cannot be read as an equivalence result because the shock era comprises only 7 editions per cup and its 95% CI is correspondingly wide (see Limitations §Third). This is the formal test of the non-monotonic time pattern noted at the descriptive level: **on the top-1 metric** the golden era spikes well above both endpoints, and the "non-monotonic across three eras" pattern rests on the well-powered early-vs-golden and golden-vs-shock contrasts (all four significant at Holm-corrected $\alpha = 0.05$; three at Holm $p < 0.01$, the fourth — empress's golden-vs-shock — at Holm $p = 2.1 \times 10^{-2}$) rather than on the underpowered early-vs-shock non-detection. On the top-3 and top-8 metrics, the pattern is **not** the same non-monotonic shape as top-1: the observed rates rise monotonically from the early era through the golden era and reach a shock-era ceiling of 100% on top-3 and top-8 (per-cup, all 7 editions × 2 cups within the shock era), so on those two thresholds the shape is early ≪ golden ≈ shock (with shock era saturating at 100%) rather than early ≈ shock ≪ golden. This is unlike top-1's non-monotonic golden-spike; the descriptive concentration is present in every threshold, but the *shape* of the era pattern differs across thresholds and the joint test at the top-3 and top-8 thresholds is correspondingly attenuated (exact p and Pearson χ^2 p both above conventional thresholds on the top-8 emperor's / empress's cells and only marginal on emperor's cup top-3). The per-cup rows are reported as the primary formal tests; both-cups pooling is not applied because the two cups within the same edition share the host prefecture, athletes, and venues and are not independent observations (Limitations §Third).

Pooled ordered logit

Table 5 reports the pooled ordered-logit specification on host rank (censored at 9 = outside top 8) with era and cup dummies as fixed effects and cluster-robust standard errors at the host-prefecture level (46 unique host prefectures across the 75-edition sample; Tokushima is the only prefecture that has never served as primary host and appears in the panel only as a co-host row of the 8th and 48th editions). The pooled specification ($n = 150$) recovers a golden-era coefficient of **−3.33** ($SE = 0.86$, $p < 0.001$; negative coefficient corresponds to better numerical rank on the ordered scale) and a shock-era coefficient of **−0.54** ($SE = 0.54$, $p = 0.317$), both relative to the early-era reference. The cup coefficient (emperor's cup vs. empress's cup reference) is −0.89 ($SE = 0.27$, $p = 0.001$), consistent with the descriptive finding that host prefectures place better on the emperor's cup than on the empress's cup at every threshold. The cup-mode-specific ordered logits (emperor's cup only $n = 75$; empress's cup only $n = 75$) both show a marked golden-era improvement (golden-era coefficients of −3.37 for emperor's cup and −3.45 for empress's cup, both $p < 0.01$), whereas the shock-era estimates differ between the two cups: emperor's cup +0.08 ($p = 0.89$, no difference was detected relative to the early-era reference) versus empress's cup −1.15 ($p = 0.045$, notably better than the early-era reference). This cup-specific divergence in the shock-era coefficient is consistent with the descriptive per-cup rates (emperor's early 56.7%, shock 42.9% — nearly identical; empress's early 30.0%, shock 42.9% — a modest increase). Only 7 of 9 potential rank cells (1, 2, 3, 4, 5, 8, and 9) are actually observed in the pooled panel; the ordered logit estimates the 6 thresholds between the observed categories, and ranks 6 and 7 are not separate categories in the estimated model.

Balmer et al. (2003)-style sport-type decomposition (2024-2025 subsidiary)

Because sport-level score data are only publicly available for the 2024 Saga and 2025 Shiga editions, the Balmer et al. (2003)-style objective/subjective decomposition is estimated as a subsidiary two-year cross-section rather than as an extension of the primary 75-completed-edition analysis. Table 6 reports the seven internally consistent point estimates that this two-year design produces (the primary raw-score specification under two inferential views in rows i-ii, plus five sensitivity variants in rows iii-vii covering log-outcome, classification, sample-inclusion, and two scale-invariant normalizations). The primary specification ($n = 6{,}991$ prefecture-sport-year-trophy cells, all 41 sports retained after handling edition-specific sport-catalog rotation between the two editions (the objective-side count changes from 17 to 16 sports across 2024→2025 because clay-target shooting was appended in 2024 and removed in 2025; the subjective- and semi-subjective-side counts are unchanged across the two editions); 47 prefecture clusters with 2 treated) recovers a host main effect of **+16.60 points per sport** ($SE = 1.25$, $p < 10^{-30}$) and a **host × subjective interaction of +16.68 points** ($SE = 7.72$, cluster-robust **$p = 0.031$**). Because the two-year window provides only 2 treated prefecture clusters (Saga in 2024 and Shiga in 2025) out of 47, the cluster-robust variance formula is known to be anti-conservative in this regime (Cameron & Miller, 2015[13]; MacKinnon & Webb, 2017[15]). We therefore accompany the cluster-robust p with a wild-cluster bootstrap (Rademacher weights, restricted null β_HS = 0, B = 999 iterations; Cameron, Gelbach & Miller, 2008[14]), which returns **bootstrap p = 0.155** and a pivotal 95% CI of [−0.42, +33.86]. Under the wild-cluster bootstrap, the interaction does not attain conventional significance, so we frame the +16.68 point estimate as directional (positive) evidence for the Balmer et al. (2003) decomposition rather than as a decisive rejection of the null.

Four internal robustness checks on the same 2024-2025 cross-section corroborate the direction of the primary interaction estimate. First, on a log-linear scale (dependent variable log(1 + score), inclusive n = 6,991), the interaction remains positive (**+0.293**, SE = 0.170, cluster-robust p = 0.084), directionally aligned with the primary linear specification (Table 6 row iii). Second, a classification-variant sensitivity re-estimates the primary specification under a stricter delimitation of the subjective category. The “combat-excluded” variant, in which subjectivity is narrowed to pure artistic scoring (gymnastics and equestrian only; 9 combat sports and 13 semi-subjective/team sports both excluded from the analysis frame rather than reclassified as objective; complementary pure-judged and combat-to-semi variants documented in the analysis code give β_HS = +18.99 and +26.40, respectively), yields **β_HS = +31.81** (SE = 8.20; cluster-robust $p < 10^{-3}$; wild-cluster bootstrap p = 0.174 on 2 treated clusters, close to and slightly larger than the primary specification’s bootstrap p = 0.155) — Table 6 row iv. Third, the zero-imputed sensitivity that treats the 106 non-participation cells as score = 0 rather than as complete-case missing gives β_HS = +16.49 (SE = 7.69, cluster-robust p = 0.032, bootstrap p = 0.154 with all 999 replicates converged) — Table 6 row v — confirming that the primary interaction direction is not driven by non-participation-cell handling. Fourth, two scale-invariant normalized outcomes tested whether the raw-score interaction is a sport-specific-ceiling artifact (see Limitations §Fifth): the z-score within sport-year-cup cell (row vi) preserves the positive direction with β_HS = +1.075 (SE = 0.359, cluster-robust p = 0.003, bootstrap p = 0.158, CI [−0.024, +2.172]) — the bootstrap p essentially matching the raw-score row ii bootstrap p = 0.155 and arguing against the result being solely a raw-unit artifact, although the magnitude depends on the normalization — while the percentile rank within the same cell (row vii) attenuates to β_HS = +0.062 (SE = 0.047, cluster-robust p = 0.190, bootstrap p = 0.498, CI [−0.015, +0.139]), indicating that the interaction is weaker under the rank scale than under the raw-score or z-score scales. Rows (i)-(vi) all preserve β_HS > 0; row (vii)’s β_HS is small and positive but its bootstrap-p is well above conventional thresholds. Taken together, several conventional cluster-robust specifications yielded $p < 0.05$ (rows i, iv, v, vi = four of the seven), but none of the reported wild-cluster bootstrap tests reached conventional significance (the primary bootstrap p sits at 0.155 in row ii, the combat-excluded bootstrap p at 0.174 in row iv, the zero-imputed bootstrap p at 0.154 in row v, and the z-score bootstrap p at 0.158 in row vi; the percentile-rank row vii attenuates further to bootstrap p = 0.498). The point estimates preserve a positive direction across rows (i)-(vi) but the inference is not decisive under the two-treated-cluster identification base — the normalized outcomes produce mixed evidence: the z-score estimate remains positive with an inferential profile matching the raw-score row, whereas the percentile-rank estimate is substantially attenuated.

Descriptively, the raw host-versus-nonhost per-sport score gap increases monotonically across the three Balmer et al. (2003) sport categories in exactly the direction the decomposition predicts (Table 7; Figure 3): objective sports show a gap of **+17.60 points** (host mean 43.63 vs. non-host mean 26.03, n_host = 65 cells across 17 sports), semi-subjective / team sports **+26.18 points** (47.03 vs. 20.85, n_host = 48 cells across 13 sports), and subjective sports **+37.91 points** (57.24 vs. 19.33, n_host = 38 cells across 11 sports). The subjective host boost of +37.9 raw points is numerically larger than the objective category's +17.6; the raw-score ratio of 2.15 must be read as descriptive rather than as evidence of a proportionally doubled bonus because per-sport raw scores are not normalized for sport-specific total-score ceilings, participant-count differences, or scoring-scale differences (see Limitations §Fifth), and the subjective-sport basket may contain sports with systematically higher total-score ceilings than the objective basket. Figure 3 displays the three category-specific host boosts with 95% confidence intervals from per-category `score ~ is_host` ordinary least squares (OLS) with prefecture-clustered SEs (47 clusters).

Cross-national context

Table 8 places the Kokutai host bonus in cross-national context. Among the four cases summarized, Japan is the clearest example in which a large host bonus (host top-1 rate 76.0% on the emperor's cup pooled across 75 editions vs. a k / 47 chance baseline of 2.13%) emerges *without* any institutional score bonus in JSPO scoring rules — in contrast to Korea's KSOC national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city[8], and in contrast to the ~45% attenuation of the Olympic host coefficient once population and per-capita GDP are added as country-level controls (Csurilla & Fertő, 2023)[9]. The Chinese National Games host effect (Lan & Yu, 2012)[11] is characterized qualitatively rather than as an explicit institutional bonus, but the five descriptive attributes Lan & Yu list (high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness) describe the same empirical phenomenon we observe at the Kokutai in more diffuse language.

Discussion

The Balmer et al. (2003) decomposition is directionally supported at the Kokutai

One exploratory finding of this study, on the 2024-2025 two-year cross-section, is that the per-sport raw-score host-versus-non-host difference is numerically larger in subjectively judged sports than in objectively measured ones, in the direction that the Balmer et al. (2003) decomposition predicts. In the 2024-2025 stacked cross-section, the primary specification (n = 6,991 inclusive prefecture-sport-year-trophy cells, all 41 sports retained after handling edition-specific sport-catalog rotation between the two editions (the objective-side count changes from 17 to 16 sports across 2024→2025 because clay-target shooting was appended in 2024 and removed in 2025; the subjective- and semi-subjective-side counts are unchanged across the two editions); 47 prefecture clusters with 2 treated) yields a host main effect of +16.60 raw score points per sport and a host × subjective interaction of +16.68 raw score points (cluster-robust $p = 0.031$; wild-cluster bootstrap $p = 0.155$ under the few-treated-clusters correction on 2 treated prefectures out of 47). The raw descriptive gap orders the three sport categories monotonically as +37.9 (subjective) > +26.2 (semi-subjective) > +17.6 (objective) — precisely the ordering that Balmer, Nevill & Williams (2003) documented at the summer Olympics. Because per-sport raw scores are not normalized for sport-specific total-score ceilings, participant-count differences, or scoring-scale differences (see Limitations §Fifth), we do not claim a proportional doubling of the underlying bonus. Two scale-invariant normalizations reported as Table 6 rows (vi)-(vii) test how the interaction behaves under alternative outcome scales: the z-score within sport-year-cup cell (row vi) preserves the positive direction with bootstrap $p = 0.158$, essentially matching the raw-score bootstrap p (0.155), whereas the percentile rank within the same cell (row vii) attenuates to bootstrap $p = 0.498$. Point-estimate direction is therefore robust to scale-invariant rescaling, but the inferential strength is attenuated on the rank scale. Given that the wild-cluster bootstrap does not reach conventional significance under the two-treated-cluster structure of this cross-section, we frame this as a *directionally* consistent first quantitative anchor for the Balmer et al. (2003) decomposition on Kokutai data — supported concordantly by (a) the raw descriptive monotonicity, (b) the log-outcome interaction direction, and (c) the linear interaction point estimate — rather than a decisive statistical confirmation. The Cameron-Gelbach-Miller (2008) pivotal wild-cluster bootstrap 95% confidence interval of [−0.42, +33.86] straddles zero and is not statistically significant at the 5% level, consistent with the bootstrap p reported in Table 6 (0.155); the point estimate is positive but the bootstrap interval includes zero. It nevertheless closes the specific gap left by Funahashi, Hibino, Ishiguro & Mano (2016), whose 2003-2011 panel documented a robust overall host bonus but did not disaggregate by sport type.

A three-variant classification sensitivity analysis further supports the invariance of the sport-type gradient to reasonable classification choices (Table 6). The combat-excluded variant — which restricts the subjective category to pure artistic scoring (体操 gymnastics + 馬術 equestrian, dropping the nine combat sports from the analysis frame entirely rather than reclassifying them as objective, and additionally dropping the 13 semi-subjective/team sports to isolate a pure objective-vs-subjective contrast (see `src/analysis_cross_section_2024_2025.py::run_bootstrap_by_variant` L553-554 for the exact filter chain and Table 6 row (iv) footnote for the sample-size accounting) — yields the largest cluster-robust point estimate on this data (β_HS = **+31.81**, SE = 8.20, cluster-robust **$p < 0.001$**; wild-cluster bootstrap $p = 0.174$, close to and slightly larger than the primary specification's bootstrap $p = 0.155$ under the few-treated-clusters correction applied consistently across all four variants), which is consistent with a pattern in which host bias appears strongest under pure artistic scoring rather than under combat-referee judgment (though this cross-variant comparison is a directional observation, not a formal test, since the combat-excluded variant's bootstrap $p = 0.174$ remains above conventional significance thresholds under the same few-treated-clusters correction). The complementary pure-judged and combat-to-semi variants (documented in the analysis code) yield β_HS = +18.99 and +26.40, respectively; the primary inclusive specification's β_HS = +16.68 together with the three classification variants (+18.99, +26.40, +31.81) preserve β_HS > 0 across the range +16.68 to +31.81, so the qualitative directional finding is invariant to how the subjective category is delimited within the plausible range spanned by Balmer et al. (2003)-consistent classification schemes.

The finding is consistent with, but does not require, a judging-bias mechanism. The Balmer et al. (2003) decomposition is an *implicit* test — an interaction of the observed type is what a judging-bias mechanism would produce, but a purely crowd-effect mechanism that is stronger in enclosed, spectator-dense subjective-sport venues (typical: judo, kendo, gymnastics halls) than in open-air objective-sport venues (typical: track and field, cycling road courses) would produce the same pattern. Without a judge-level dataset we cannot uniquely identify judging bias as the mechanism. What we can say is that a purely uniform material mechanism would be less likely to produce this sport-type gradient, although sport-specific preparation, venue familiarity, and transfer-athlete channels cannot be ruled out. The interaction we recover is compatible with — but not preferentially supportive of — a crowd/judging pathway. Nomura (2022)[10] provides collateral evidence from Japanese J1 football that home advantage disappeared during the reduced-occupancy 2020 COVID-19 season and — through multiple-group structural equation modeling — that the crowd's influence on referees' decisions vanished under the same conditions, further supporting the crowd–referee pathway. This analogy from J1 to the Kokutai is inherently indirect: the two Kokutai editions that would have been the natural in-domain analogues of Nomura's reduced-occupancy 2020 J1 season — the 75th (2020 Kagoshima) and 76th (2021 Mie) editions — were cancelled for COVID-19 (see Methods and Limitations), so the crowd-referee mechanism cannot be tested at the Kokutai directly and the J1 evidence is offered as suggestive rather than confirmatory for Kokutai-specific inference.

Two host-loss clusters, not one

The popular framing of a "post-2016 Tokyo winning streak" is factually incorrect. Host prefectures won both 2018 Fukui (2,896 vs. Tokyo's 2,246) and 2019 Ibaraki (2,569 vs. Tokyo's 2,217) under exactly the same national scoring rules that produced the Tokyo wins in 2016 Iwate and 2017 Ehime. The 2020 and 2021 editions were cancelled for COVID-19, with the 2020 Kagoshima program re-scheduled as a special edition in 2023. Within the 2016-2025 shock era (7 non-cancelled editions × 2 cups = 14 host-cup cells), the host top-1 rate is 42.9% (6 of 14; both cups pooled), reflecting host losses concentrated in 2016 Iwate, 2017 Ehime, 2022 Tochigi, and 2024 Saga, interspersed with host wins in 2018 Fukui, 2019 Ibaraki, and 2025 Shiga rather than a uniform post-2016 host-loss regime. The 2002 Kōchi observation — the sole edition on which the primary host finished outside the top 8 (rank > 8) outside the early era, in the middle of the 1978-2015 golden era — sits apart from these clusters and is documented separately in Figure 1. The within-shock-era heterogeneity — 2016 Iwate / 2017 Ehime / 2022 Tochigi / 2024 Saga host losses interspersed with 2018 Fukui / 2019 Ibaraki / 2025 Shiga host wins under identical scoring rules — suggests a modeling recommendation for future Japanese work on the Kokutai host effect: 2016 should not be treated as a single structural break, and a framing that preserves the interspersed 2018/2019/2025 host wins within the broader 2016-2025 shock era is likely to be more informative than a single-break design.

East-Asian institutional context

The Kokutai host bonus emerges *in the absence of* an explicit institutional host bonus in JSPO scoring rules (Table 8 summarizes the cross-national institutional context). This stands in contrast to the Korean Sport & Olympic Committee's national comprehensive sports competition, which explicitly adds a 20% score bonus to the host city (10% from 2001, raised to 20% in 2010),[8] and to the qualitative characterization Lan and Yu (2012)[11] provide of the Chinese National Games host effect, whose five listed characteristics — high efficacy, gradual increase, delayed effect, exclusivity, and double-edgedness — describe roughly the same phenomenon we observe empirically at the Kokutai but in more diffuse qualitative language (this reference is retained at a *Low confidence* level: the source is a Chinese-local journal not indexed on Crossref, and only the abstract-level Chinese-language summary was accessible to us via a secondary-source portal; independent verification of the five-characteristics enumeration through the primary text was not possible with tools available to us, and the reference should be read in the "descriptive parallel" register rather than as a substantive citation). The present study is arguably an implicit decomposition of Lan & Yu's "exclusivity" and "double-edgedness" characteristics: exclusivity because the host bonus is concentrated in a subset of sports (subjectively judged), and double-edgedness because a large host bonus in a subjective sport increases both the host's expected medal count and the perceived legitimacy risk of that medal (see the Suetsugu 2024/2025 qualitative critique below).

Overlap and boundary with Suetsugu (2024, 2025)

The Suetsugu papers[4,5] provide a karate-specific qualitative critique of Kokutai outcomes, using the "host-victory-first mindset" framing (a phrase corresponding to the "開催地優勝至上主義" title of Suetsugu 2025) in a single-sport context. Our whole-sport panel-quantitative decomposition is complementary rather than overlapping: Suetsugu's papers focus on a single subjectively judged sport (karate) and characterize the outcome pattern in that sport in normative terms; our analysis provides exploratory quantitative evidence, on the 2024-2025 cross-section only, consistent with a sport-type pattern (per-sport host boost larger in subjectively judged sports than in objectively measured ones) rather than confining the phenomenon to karate; the two-treated-cluster identification base and the wild-cluster bootstrap $p = 0.155$ mean this evidence is directional rather than confirmatory. The two contributions can stand side by side: Suetsugu characterizes the qualitative concern within a single sport; we provide quantitative evidence consistent with the same interaction pattern across all sports.

We deliberately avoid the normative language Suetsugu uses. The observed interaction is consistent with judging bias, but it is also consistent with subject-sport-specific crowd effects, subject-sport-specific practice-environment effects, and subject-sport-specific transfer-athlete effects. The present study does not adjudicate among these mechanisms; it provides descriptive evidence that the raw per-sport score gap is numerically larger in subjectively judged sports than in objectively measured ones, in the direction of the Balmer et al. (2003) prediction (see Limitations §Fifth for the reasons the raw-score ratio should not be read as a proportional bonus).

The identified interaction is therefore a reduced-form estimate that lumps judge-bias, crowd-effect, practice-environment, and transfer-athlete channels; the absence of a JSPO judge-of-record roster (see Limitations, Second) prevents attribution of the observed effect to any single channel. A judging-bias mechanism, if present, would plausibly operate through host prefectures that supply larger judge cohorts for subjectively judged sports (gymnastics, kendo, karate, etc.) than for objectively measured sports, but we have no data with which to verify or quantify this asymmetry. The reduced-form estimate reported here should therefore not be read as identifying judge-bias per se; it is a reduced-form association that reflects an unidentified mixture of these candidate channels.

Robustness across specifications

The reality-test finding — that the host prefecture reaches the top-1, top-3, and top-8 at rates far exceeding the *k* / 47 chance baseline — replicates in every era × cup × threshold cell of Table 2 (see Results), and the 46-versus-47-state sensitivity check confirms that the excess above chance differs from the 47-state baseline by less than 0.005 across every early-era threshold-cup cell. Unlike the ≈45% attenuation that Csurilla & Fertő (2023)[9] report for the country-level Olympic host coefficient when population and per-capita GDP are added as controls, the Kokutai host bonus emerges *without* any institutional score bonus at the primary-host level (Table 8), consistent with the Kokutai host-prefecture rotation being fixed by JSPO decades in advance rather than an economically endogenous selection.

Future work

A companion paper (in preparation) extends the categorical framework of the present study by introducing a continuous subjectivity gradient — scored on a Balmer et al. (2003)-consistent rubric across the full 40+ Kokutai sport catalogue with inter-rater reliability reported — and a sport-level exploratory screen for host-overperformance using newly collected competition-level top-N placement data. The companion paper is intended to complement (rather than supersede) the present categorical decomposition: the categorical contrast established here provides the qualitative direction, and the gradient/screen analyses would provide a finer-grained mechanistic decomposition together with case-study treatments of specific borderline sports (剣道 kendo, 柔道 judo, 相撲 sumo) that the current three-category framework treats uniformly.

Limitations

Primary conceptual limitation (causal identification). The *k*/47 uniform-selection chance baseline used throughout the reality-test and era-heterogeneity analyses documents a descriptive overrepresentation of host prefectures among top-*k* finishers rather than identifying a causal home-advantage effect. The counterfactual a causal claim would require — the same prefecture's performance with and without hosting, holding fixed prefecture-level competitive strength, population, budget, JSPO strengthening funding, and pre-hosting athlete-development trajectories — is not identified by the *k*/47 baseline, and the permutation simulation shares the same exchangeability assumption. The observed top-*k* concentration is consistent with a mixture of (a) a genuine causal hosting effect (venue familiarity, crowd support, judging bias), (b) systematic correlation between the host role and prefecture-level competitive strength under the JSPO rotational schedule, and (c) pre-hosting selection effects via "強化" (*kyōka*, athlete-strengthening) budgets that host prefectures deploy ahead of their host year. Proper causal identification would require a within-prefecture event-study on a continuous outcome, a socioeconomic-controls-augmented panel, a synthetic-control design, or a Zitzewitz (2006)-style judge-level analysis; none is implemented here. The descriptive framing of the Abstract, Introduction, Results, and Conclusion reflects this conceptual limitation. The six numbered limitations below concern specific data and inference features of the analyses actually reported.

First — Sample and data availability. Edition-level sport-by-prefecture score data are only publicly available for the 2024 Saga and 2025 Shiga editions (earlier editions publish only prefecture-level overall standings), restricting the Balmer-style interaction test to a two-year cross-section with only 2 treated prefecture clusters out of 47 (this identifying structure is discussed in §Fourth below). The Cabinet Office ESRI 2008SNA prefectural GDP series that would anchor a socioeconomic-controls-augmented panel is available only through fiscal 2022; extension to 2023-2024 awaits future ESRI releases. The JSPO official retrospective volume `01_kokutai.pdf` (692 pages covering editions 1-65) contains competition-level top-6 placements but was not incorporated because of a three-column layout that fragments same-competition rows, top-6 (not top-8) as the volume-wide standard, frequent tie placements (multiple ⑤ entries), individual-event affiliations reported as clubs/schools/corporate teams rather than prefectures, and format drift across the 65-edition window. Pre-2024 subjective-sport score data are patchy (kendo published a per-prefecture score table only for the 2016 Iwate 71st edition and discontinued it by 2022 Tochigi 77th). The 80th edition (2026 Aomori) was not yet held at manuscript preparation; if the 47-prefecture overall standings are published before submission the host-rank pipeline can be extended by one edition, otherwise the analysis remains fixed at the 75 completed editions. All sample sizes reported (n = 150 host-rank panel; n = 6,991 primary cross-section) are availability-driven from current JSPO and ESRI releases rather than *ex ante* power-calculated. The Suetsugu (2024, 2025) full texts could not be reliably extracted from the Komazawa University institutional repository's WEKO3 octet-stream downloads; citations are at bibliographic-plus-CiNii-link level.

Second — Mechanism identification. We could not obtain the JSPO judge-of-record roster (name + prefecture of affiliation) for any Kokutai edition despite four independent search paths (JSPO regulations catalog, seven individual-edition sampling, kendo-specific archives, general web search). The Balmer-style decomposition is therefore the only implicit test of judging bias we can offer; a direct Zitzewitz (2006)-style judge-level analysis is not available for Kokutai data at present. The reduced-form interaction we recover in the 2024-2025 cross-section lumps judge-bias, crowd-effect, practice-environment, and transfer-athlete channels and cannot attribute the observed magnitude to any single mechanism.

Third — Baseline and boundary choices for the 75-edition analysis. Four choices affect the reported inference on the 75-edition panel: (i) the $k/47$ null uses the current 47-prefecture composition, whereas 24 of the 30 early-era editions (pre-1972 Okinawa reversion) had 46 prefectures — under a conservative 46-

state null applied to the full early era, the excess above chance differs from the 47-state baseline by less than 0.005 across all threshold-cup cells (Supplementary Materials §S3); (ii) the Table 2 and Table 4 statistics are reported per-cup because the two cups within the same edition share the host prefecture, athletes, and venues — a pooled $n = 150$ test would violate independence and understate p-values (an edition-clustered GEE or random-intercept respecification is left as future work); (iii) the top-1 early-vs-shock pairwise Fisher tests do not detect a difference on either cup (Holm-corrected $p = 1.00$), but this reflects the shock era's small sample ($n = 7$ per cup; Wilson 95% CI [0.099, 0.816] for 3/7) rather than positive equivalence — a proper TOST equivalence test against a pre-specified margin would be required to claim equivalence rigorously; (iv) the 1978 golden-era and 2016 shock-era boundaries were selected with knowledge of the outcome data (author-defined, not pre-registered). A 30-combination boundary-sensitivity grid (golden-start $\in$ {1975..1980} × shock-start $\in$ {2014..2018} × 2 cups; Supplementary Materials §S4) confirms that the golden > early and golden > shock ordering and per-cup exact $p < 0.05$ survive every considered boundary shift.

Fourth — Statistical specification choices for the ordered logit and cross-section. Three specification choices bear on the reported inference: (i) no formal proportional-odds diagnostic (e.g., a Brant-style test) was performed for any ordered-logit specification reported in this paper (the v3 pooled ordered logit in Table 5, $n = 150$, or the per-cup ordered logits, $n = 75$ each) — the ordered-logit results are therefore reported as secondary cross-checks, and the substantive era-heterogeneity conclusions rest primarily on the conditional Pearson χ^2 exact tests (Table 4) and the pairwise Fisher exact tests with Holm-Bonferroni correction, neither of which depends on the PO assumption; the top-*k* concentration conclusions rest on the one-sample exact binomial tests and the permutation simulation (Tables 2-3), which also do not invoke the PO assumption; (ii) the β_HS interaction in the 2024-2025 cross-section is identified off the difference between Saga and Shiga in their subjective-vs-objective host-cell contrasts — this is a specific 2-prefecture design in which the interaction's variance is a function of exactly how those two prefectures happened to differ in subjective-sport versus objective-sport performance, which is one of the mechanical reasons the wild-cluster bootstrap p (0.155) diverges from the naive cluster-robust p (0.031); (iii) the wild-cluster bootstrap uses Rademacher weights, and with only two treated clusters the discrete Rademacher distribution places mass at just four sign patterns — the Mammen (1993) two-point-discrete-asymmetric weight sensitivity check (Supplementary Materials §S1) returns $p = 0.181$, qualitatively identical to Rademacher's 0.155, so the reported non-significance is not driven by Rademacher-weight discreteness. Extending the interaction test to a longer panel with more host prefectures would relax both the anti-conservative-SE concern and the two-prefecture asymmetric-identification concern.

Fifth — Cross-section outcome and classification choices. Three choices affect the reported cross-section results: (i) the primary specification uses the raw per-sport score `score` as the outcome, and sport fixed effects absorb sport-specific mean levels but not sport-specific total-score ceilings, participant-count differences, or scoring-scale differences — a host boost in a high-total-score sport can be numerically larger than in a low-total-score sport at the same underlying advantage rate; we address this partially with the log-outcome (Table 6 row iii) and directly with the z-score-within-sport-year-cup and percentile-rank-within-sport-year-cup normalized specifications (Table 6 rows vi-vii); the z-score row preserves the positive direction and matches the raw-score bootstrap p (0.158 vs. 0.155), while the percentile-rank row attenuates to bootstrap p = 0.498, an honest mixed-evidence outcome across scale-invariant transformations; (ii) the tripartite subjective / semi-subjective / objective sport classification follows Balmer et al. (2003) but the subjective basket spans heterogeneous scoring paradigms (artistic scoring, combat referee scoring, combat victory decisions, traditional-format scoring); the three-variant classification sensitivity (inclusive, pure-judged, combat-excluded) preserves $\beta_{HS} > 0$ across +16.68 to +31.81 but does not resolve the theoretical concern that the binary indicator groups conceptually distinct mechanisms — a finer decomposition would fragment the two-prefecture identifying variance beyond usable resolution; (iii) 106 of the theoretical 7,097 prefecture-sport-year cells are absent from JSPO-published data (non-participation), and the primary Table 6 specification treats these as complete-case missing (n = 6,991); a zero-imputed sensitivity (Table 6 row v) that treats non-participation as score = 0 returns β_{HS} = +16.49 (vs. complete-case +16.68, difference −0.19 ≈ 1.1%) with all 999 bootstrap replicates converged and bootstrap p = 0.154, confirming that the primary interaction direction is not driven by non-participation-cell handling. No formal outlier-handling procedure (winsorization, trimming, influence diagnostics) is applied to the continuous per-sport score outcome; sensitivity to individual host-sport-year cells with unusually large boosts is not separately quantified.

Conclusion

Across the 75 completed editions in the 3-79 span (1948-2025), host prefectures were substantially overrepresented among top-*k* finishers relative to a *k*/47 uniform-selection chance baseline: on both the emperor's cup and the empress's cup, the host reached the top-8 in 70 of 75 editions (93.3% each; against 17.02% chance); the host reached the top-3 in 69 of 75 emperor's-cup editions (92.0%) and 64 of 75 empress's-cup editions (85.3%; against 6.38% chance); and the host won the championship in 57 of 75 emperor's-cup editions (76.0%) and 47 of 75 empress's-cup editions (62.7%; against 2.13% chance). This descriptive top-*k* concentration on the top-1 metric was highest during 1978-2015 (emperor's cup 37 of 38 = 97.4%, empress's cup 35 of 38 = 92.1%) and lower during the pre-1978 formative window (emperor's cup 17 of 30 = 56.7%, empress's cup 9 of 30 = 30.0%) and the post-2016 shock window (3 of 7 = 42.9% on both cups); on the top-1 metric the temporal pattern is therefore non-monotonic on both cups across the three author-defined era periods, whereas on the top-3 and top-8 metrics the observed rates rise monotonically through the golden era and reach a 100% ceiling in the shock era (early < golden ≈ shock at ceiling). The precise timing and the causal mechanisms of the top-1 non-monotonic pattern remain uncertain: the era boundaries themselves (1978 as the golden-era onset and 2016 as the shock-era onset) were selected with knowledge of the outcome data rather than pre-registered, the shock era comprises only 7 non-cancelled editions (which limits the precision of any post-2016 comparison and precludes reading the per-cup early-vs-shock Fisher p ≈ 0.68 / 0.66 as an equivalence result), and the *k*/47 baseline treats the host role as if drawn uniformly at random from the 47 prefectures — it therefore documents a concentration pattern rather than a causal home-advantage effect that would require holding fixed prefecture-level competitive strength, population, budget, and pre-hosting athlete development. This descriptive concentration is documented by the Funahashi et al. (2016) 2003-2011 panel at the aggregate score level (Funahashi-specification replication retained in Supplementary Materials §S2 for pipeline validation), and stands in institutional contrast to the Korean KSOC's explicit 20% host multiplier and in qualitative contrast to Csurilla & Fertő's (2023) finding that the Olympic host coefficient attenuates once population and per-capita GDP controls are introduced.

An exploratory analysis of the 2024 Saga and 2025 Shiga editions — the only two for which JSPO publicly releases the full 47-prefecture × ~40-sport disaggregation — suggested larger per-sport host-versus-non-host score differences in subjectively classified sports than in objectively measured ones, in the direction that Balmer et al. (2003) predicts: the descriptive per-sport host boost runs monotonically from +37.9 points in the subjective category through +26.2 in semi-subjective/team to +17.6 in objective (Table 7), and the regression host × subjective interaction is +16.68 points (cluster-robust $p = 0.031$). This estimate, however, rests on only two treated prefectures out of 47, is sensitive to the choice of inference method (wild-cluster bootstrap $p = 0.155$ under the appropriate few-treated-clusters correction; Mammen-weighted bootstrap $p = 0.181$; Cameron-Gelbach-Miller pivotal 95% CI [−0.42, +33.86]), and does not identify judging bias, subject-sport-specific crowd effects, subject-sport-specific practice-environment effects, or any other specific causal mechanism. It also depends on the point-estimate stability of per-sport raw scores as the outcome variable — a choice that does not normalize across sport-specific total-score ceilings, participant-count differences, or scoring-scale differences (see Limitations §Fifth). A three-variant classification sensitivity preserves $\beta_{HS} > 0$ across the range +16.68 (primary inclusive) to +31.81 (combat-excluded pure-judged) with bootstrap p values in the 0.155-0.65 range across variants, but all variants share the same two-treated-prefecture limitation and cannot be read as independent identification of a causal sport-type effect. Two scale-invariant normalized outcomes (Table 6 rows vi-vii) produce mixed evidence on the sport-type interaction: the z-score within sport-year-cup cell preserves the positive direction with bootstrap $p = 0.158$ — matching the raw-score row (ii) bootstrap $p = 0.155$ and arguing against the result being solely a raw-unit artifact, although the magnitude depends on the normalization — while the percentile rank within the same cell attenuates to bootstrap $p = 0.498$, so the interaction weakens substantially on the rank scale. The exploratory result should therefore be read as descriptive directional evidence consistent with the Balmer et al. (2003) prediction that would require broader sport-level historical data (which the JSPO archive does not currently release for editions prior to 2024) or a parallel Zitzewitz-style judge-level analysis to be sharpened into a decisive causal test.

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Author Contributions (Contributor Roles Taxonomy, CRediT)

Mizuki Shirai: Conceptualization, Methodology, Investigation, Data Curation, Formal Analysis, Software, Writing — Original Draft, Writing — Review & Editing, Visualization, Project Administration.

Conflict of Interest

The author declares no conflicts of interest per International Committee of Medical Journal Editors (ICMJE) guidelines.

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Data Availability

All input data used in this study are publicly available. Prefecture rankings are from the Nagano Prefecture Sports Association (https://www.nagano-sports.or.jp/kokutai/record/high_rank.html); individual-edition PDFs and 2024-2025 Excel breakdowns are from the JSPO official archive (https://www.japan-sports.or.jp/kokutai/tabid183.html); socioeconomic covariates are from the Cabinet Office ESRI prefectural accounts (https://www.esri.cao.go.jp/jp/sna/data/data_list/kenmin/files/contents/main_2022.html).

Analysis code (Python 3.14, pandas 3.0.5, statsmodels 0.14.6, matplotlib 3.11.1, weasyprint 69.0) — including the data-assembly scripts, all v3 regression modules (v3 pooled ordered logit on the 75-completed-edition host-rank panel (span 3-79), Funahashi replication OLS, 2024-2025 cross-section, and the full-enumeration conditional Pearson χ^2 exact test used as the primary global test in Table 4, with a Monte Carlo permutation implementation on the same statistic as a numerical cross-check), the wild-cluster bootstrap implementation, the plotting pipeline, and the PDF build pipeline — is publicly available at https://github.com/rehabilitation-collaboration/kokutai-home-advantage (MIT License). All numerical results reported in this manuscript are reproducible from the analysis code at repository commit `936b358` — this single commit provides the complete v3 analysis pipeline, including the inclusive zero-imputed sensitivity function used for Table 6 row v, the design-matrix respecification for the collinearity fix (removal of the sport-fixed-attribute main effects that are exact linear combinations of the sport FE), the full-enumeration conditional Pearson χ^2 exact test (`_conditional_exact_p_chi2`) that serves as the primary global test in Table 4, and the Monte Carlo permutation implementation (`_mc_permutation_chi2`) that serves as its numerical cross-check. This commit is publicly accessible on the GitHub repository above as the immutable reproducibility anchor for the results reported here. The deprecated 2012-2022 legacy 47-prefecture-year Csurilla-style staged-confounder implementation, the pooled ordered/binary logit implementation on that legacy panel, and the Brant-style partial-proportional-odds diagnostic implementation used for the preliminary diagnostic reported in Limitations §Fourth and for the pipeline validation reported in Supplementary Materials §S2 Extension through 2022 are archived in the git history at commit `fa07fd2` and earlier of the same GitHub repository (modules `src/analysis_main.py` and `src/analysis_confounders.py`, invoked via `run_all_models.py` in that commit). This commitment is made explicit to avoid an asymmetric-disclosure gap in which AI-usage transparency (see Acknowledgments — LLM-assisted development is fully disclosed at the point of first posting) would run ahead of code transparency and thereby impair independent reviewer verification of the LLM-assisted numerical results reported here.

Tables

Table 1. Host top-k rates by era × cup × threshold (v3 primary; n = 150 = 75 editions × 2 cups)

Era	Cup	n editions	n top-1	n top-3	n top-8	Rate top-1	Rate top-3	Rate top-8
early (1948-1977)	emperor’s	30	17	25	26	0.567	0.833	0.867
early (1948-1977)	empress’s	30	9	20	26	0.300	0.667	0.867
golden (1978-2015)	emperor’s	38	37	37	37	0.974	0.974	0.974
golden (1978-2015)	empress’s	38	35	37	37	0.921	0.974	0.974
shock (2016-2025)	emperor’s	7	3	7	7	0.429	1.000	1.000
shock (2016-2025)	empress’s	7	3	7	7	0.429	1.000	1.000
all (1948-2025)	emperor’s	75	57	69	70	**0.760**	**0.920**	**0.933**
all (1948-2025)	empress’s	75	47	64	70	**0.627**	**0.853**	**0.933**

Panel: 150 prefecture-cup observations = 75 non-cancelled editions × emperor’s cup and empress’s cup. Editions excluded: 1st (1946) and 2nd (1947) — no official 47-prefecture ranking available; 75th (2020 Kagoshima) and 76th (2021 Mie) — cancelled for COVID-19 (see Methods). Three multi-prefecture co-hosted editions (7th, 8th, 48th) are normalized to their primary host (see Methods and Data Availability). The three-era decomposition uses ERA_BOUNDARIES = {early: 1948-1977, golden: 1978-2015, shock: 2016-2025}. The non-monotonic pattern on the top-1 metric (early 0.433 pooled → golden 0.947 pooled → shock 0.429 pooled) is the descriptive fingerprint of the golden-era spike sandwiched between the early and shock eras; the pattern is formally tested in Table 4.

Table 2. One-sample proportion test: host top-k rate vs. chance baseline k/47 (per-cup primary; six confirmatory rows = 2 cups × 3 thresholds)

Cup	Threshold	n	n_success	Observed rate	Null rate (k/47)	Excess	Wilson 95% CI (two-sided)	Exact-binomial upper-tail p
emperor’s	top-1	75	57	0.760	0.021	0.739	[0.652, 0.842]	3.2×10^{-79} (< 10^{-40} safe cap)
emperor’s	top-3	75	69	0.920	0.064	0.856	[0.836, 0.963]	4.8×10^{-75} (< 10^{-40} safe cap)
emperor’s	top-8	75	70	0.933	0.170	0.763	[0.853, 0.971]	1.0×10^{-47} (< 10^{-40} safe cap)
empress’s	top-1	75	47	0.627	0.021	0.605	[0.514, 0.727]	4.5×10^{-59} (< 10^{-40} safe cap)
empress’s	top-3	75	64	0.853	0.064	0.789	[0.756, 0.916]	8.0×10^{-65} (< 10^{-40} safe cap)
empress’s	top-8	75	70	0.933	0.170	0.763	[0.853, 0.971]	1.0×10^{-47} (< 10^{-40} safe cap)

Six primary confirmatory rows = 2 cups × 3 thresholds, matching the family of six one-sample tests defined in Methods §Multiple comparisons. All six observed p-values sit far below the Bonferroni-corrected per-test threshold α_single = 0.05 / 6 ≈ 0.0083; every row survives Bonferroni correction under a family of six. Exact-binomial one-sided upper-tail p-values are computed from `scipy.stats.binomtest` *with* `alternative='greater'` *and are reported directly (to two significant figures on the exponent) with the safe-cap notation* $< 10^{-40}$ *appended, chosen to cover the least extreme of the six rows (the two top-8 rows at* $\approx 1.02 \times 10^{-47}$*). Per-cup results are reported here as the primary specification because the emperor’s cup and empress’s cup within the same edition are not independent observations (they share the host prefecture, athletes, and venues) and a naive both-cups pooled n = 150 exact binomial test would violate the independence assumption of the underlying test statistic and understate the per-test p-values (see Limitations §Third). Era-restricted per-cup breakdowns (18 additional cells: 3 eras × 3 thresholds × 2 cups; the 6 per-cup all-era rows above bring the per-cup dump to 24 rows total) are documented in* `results/analysis_main_v3.txt` *[2]. A 46-state sensitivity check (early era, pre-Okinawa reversion) is discussed in the accompanying text and Limitations; the excess above chance differs from the 47-state baseline by less than 0.005 across every threshold-cup cell (data appendix in Supplement §S3).*

Table 3. Permutation test: host_pref reassigned uniformly at random within each edition (n_perm = 10,000; Phipson & Smyth 2010 Monte Carlo lower-bound correction)

Threshold	Cup	Observed rate	Null mean	Null SD	Null 5th percentile	Null 95th percentile	p_perm
top-1	emperor’s	0.760	0.021	0.017	0.000	0.053	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$
top-3	emperor’s	0.920	0.064	0.028	0.027	0.107	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$
top-8	emperor’s	0.933	0.169	0.043	0.107	0.240	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$
top-1	empress’s	0.627	0.021	0.017	0.000	0.053	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$
top-3	empress’s	0.853	0.064	0.028	0.027	0.107	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$
top-8	empress’s	0.933	0.166	0.043	0.093	0.240	$1 / 10{,}001 \approx 9.999 \times 10^{-5}$

For each threshold × cup cell, 10,000 permutations reassign the host label uniformly at random over the 47 prefectures within each edition and recompute the observed top-k host rate under the shuffled labels. The empirical count of permutations reaching or exceeding the observed rate was zero across all six cells; the Phipson & Smyth (2010) Monte Carlo lower-bound correction `(count + 1) / (n_perm + 1)` *yields a common p-value of* 9.999×10^{-5}*. The permutation null concentrates tightly around the theoretical chance rate (null mean ≈ k/47), confirming that the shuffle preserves the fixed 47-prefecture within-edition structure without inducing extraneous variance.*

Table 4. Three-era global test (conditional Pearson χ² exact via full enumeration; Pearson χ² asymptotic as sensitivity) and Holm-corrected pairwise Fisher exact (host top-k rate by era; per-cup primary)

Cup	Thresh.	Early	Golden	Shock	Exact p (primary)	χ² p (asymp.)	Fisher early–gold. (Holm)	Fisher gold.–shock (Holm)	Fisher early–shock (Holm)
emperor's	**top-1**	**0.567**	**0.974**	**0.429**	**3.6×10^{-5}**	4.8×10^{-5}	**1.9×10^{-4}**	**4.4×10^{-3}**	1.000
emperor's	top-3	0.833	0.974	1.000	0.084	0.076	0.401	1.000	1.000
emperor's	top-8	0.867	0.974	1.000	0.204	0.162	0.972	1.000	1.000
empress's	**top-1**	**0.300**	**0.921**	**0.429**	**5.4×10^{-8}**	5.2×10^{-7}	**5.4×10^{-7}**	**2.1×10^{-2}**	1.000
empress's	top-3	0.667	0.974	1.000	1.1×10^{-3}	9.4×10^{-4}	4.7×10^{-3}	1.000	0.620
empress's	top-8	0.867	0.974	1.000	0.204	0.162	0.972	1.000	1.000

Primary global test: conditional Pearson χ² exact test obtained by fully enumerating all 2 × 3 contingency tables with fixed row and column marginals (100-600 candidate tables per cell) and summing the multivariate hypergeometric probabilities of tables whose Pearson χ² equals or exceeds the observed statistic. This is exact rather than Monte-Carlo-approximated and avoids the shock-era expected-count violation of the standard Pearson χ² asymptotic (empress's cup top-1 shock cell: expected non-success ≈ 2.61, expected success ≈ 4.39). A Monte Carlo permutation p on the same statistic (n_perm = 10,000) is computed as a numerical cross-check and matches the enumerated exact p to within 2×10^{-2} on every row (dumped in `results/analysis_main_v3.txt` *[4]). Pairwise Fisher contrasts are Holm-Bonferroni corrected across the six per-cup contrasts (2 cups × 3 era-pairs) using* `statsmodels.stats.multitest.multipletests(method='holm')`*; the reported Holm-corrected p is not less than the raw Fisher p, and all four golden-era contrasts on the top-1 metric survive Holm correction (raw Fisher p reported in* `results/analysis_main_v3.txt` *[4b]). The two per-cup top-1 rows (highlighted) are the primary formal tests of the non-monotonic time pattern. On the top-3 and top-8 thresholds the observed rates rise monotonically from the early era through the golden era and reach a 100% ceiling in the shock era (early < golden ≈ shock at the top-3/top-8 ceiling) rather than repeating the top-1 non-monotonic pattern; the joint tests are correspondingly attenuated (top-3 empress's cup remains significant; top-8 both cups is not detected). The per-cup rows are the primary formal tests; a both-cups-pooled row is not reported because the two cups within the same edition are not independent observations (see Limitations §Third). Fisher-exact p-values reported to 2 significant figures.*

Table 5. Pooled ordered logit (rank_ordinal ~ C(era) + C(cup); cluster-robust SE at host_pref_code, 46 unique host prefectures)

Cup mode	Variable	Coefficient	SE	z	p	95% CI (Wald)	n_obs
pooled (both cups)	era_golden (vs. early ref)	**−3.332**	0.862	−3.86	**1.1×10^{-4}**	[−5.02, −1.64]	150
pooled	era_shock (vs. early ref)	−0.540	0.540	−1.00	0.317	[−1.60, +0.52]	150
pooled	cup_tennou (vs. kougou ref)	−0.892	0.270	−3.30	**9.6×10^{-4}**	[−1.42, −0.36]	150
emperor’s cup only	era_golden	−3.368	1.201	−2.81	**5.0×10^{-3}**	[−5.72, −1.01]	75
emperor’s cup only	era_shock	+0.081	0.582	+0.14	0.890	[−1.06, +1.22]	75
empress’s cup only	era_golden	−3.449	0.881	−3.92	**8.8×10^{-5}**	[−5.18, −1.72]	75
empress’s cup only	era_shock	−1.153	0.574	−2.01	**0.045**	[−2.28, −0.03]	75

Reference categories: era = early (1948-1977) and cup = kougou (empress’s cup). Negative era coefficient corresponds to better (numerically smaller) host rank on the ordered scale. 95% Wald confidence intervals are computed as coefficient ± 1.96 × cluster-robust SE. The two cup-mode-specific rows (emperor’s cup only and empress’s cup only) are the primary specifications; the pooled (both cups) row is retained here for descriptive comparison but the pooled specification’s cluster-robust SE is clustered only at `host_pref_code` *and does not additionally cluster at the edition level to account for the within-edition correlation of the two cups (which share the host prefecture, athletes, and venues; see Limitations §Third). An edition-clustered or two-way (edition × prefecture) clustered ordered logit is left as future work; the pooled row’s SE is therefore likely understated for the era-coefficient inferences on the pooled specification, and the per-cup rows should be treated as the primary formal estimates. Cluster-robust standard errors are computed at* `host_pref_code` *(46 unique clusters; Tokushima is the only prefecture that has never served as primary host and appears only as a co-host row of the 8th and 48th editions). Threshold parameters (cut-points between adjacent observed ranks) are omitted for brevity; the observed ranks span only 7 of 9 latent categories (rank 6 and rank 7 are unobserved across all 150 rows), so the threshold spans between observed rank 5 and observed rank 8 encapsulate three latent thresholds that the ordered logit does not separately identify. Full model output (including thresholds) is documented in* `results/analysis_main_v3.txt` *[5]; the log-likelihoods are −126.15 (pooled), −48.19 (emperor’s cup), and −74.29 (empress’s cup).*

Table 6. 2024-2025 cross-section: Balmer et al. (2003)-style sport-type decomposition (2 treated prefectures out of 47; seven internally consistent point estimates)

Row	Specification (β_HS reported)	n	β_H	β_HS	SE	Cluster-robust p	Bootstrap p	95% CI
(i)	Primary inclusive (raw score), cluster-robust view	6,991	+16.60	**+16.68**	7.72	**0.031**	—	[+1.55, +31.81] cl.
(ii)	Primary inclusive (raw score), wild-cluster bootstrap view	6,991	+16.60	**+16.68**	7.72	0.031	**0.155**	[−0.42, +33.86] boot.
(iii)	Log-outcome robustness: log(1 + score), inclusive	6,991	+0.467	+0.293	0.170	0.084	—	[−0.040, +0.626] cl.
(iv)	Combat-excluded variant (体操 + 馬術 only vs. objective)	3,359	+10.71	**+31.81**	8.20	**1.0×10^{-4}**	**0.174**	n.c.
(v)	Zero-imputed sensitivity (106 non-participation cells → 0)	7,097	+16.65	**+16.49**	7.69	0.032	**0.154**	[−0.40, +33.43] boot.
(vi)	Normalized z-score within sport-year-cup cell	6,991	+0.781	**+1.075**	0.359	0.003	**0.158**	[−0.024, +2.172] boot.
(vii)	Normalized percentile rank within sport-year-cup cell	6,991	+0.193	**+0.062**	0.047	0.190	0.498	[−0.015, +0.139] boot.

Model: `score_isc = α + β_H · host_isc + β_HS · (host × subj)_isc + η_i + κ_c + v_t + ξ_p + ϵ_isc` *with prefecture (η_i), sport (κ_c), year (v_t), and trophy (ξ_p) fixed effects; 47 prefecture clusters (Saga in 2024 and Shiga in 2025 are the two treated). The sport-fixed-attribute main effect β_S · subj_c is omitted because subj_c is a sport-level constant and therefore an exact linear combination of the sport fixed effects κ_c; only the host × subjective interaction is identified. Rows (i)-(ii) are the same primary raw-score specification under two inferential views (cluster-robust and wild-cluster bootstrap Rademacher B = 999 restricted-null); the point estimates are identical. Rows (iii)-(vii) are five sensitivity specifications: (iii) log-outcome direction check; (iv) combat-excluded classification variant (subjectivity narrowed to 体操 + 馬術 only; nine combat sports and thirteen semi-subjective/team sports both dropped from the analysis frame; the largest point estimate on this data); (v) zero-imputed sample-inclusion sensitivity (the 106 non-participation cells imputed to score = 0 rather than dropped); (vi) z-score normalized within sport-year-cup cells (scale-invariant, addresses sport-specific ceiling and participant-count differences — see Limitations §Fifth); (vii) percentile rank within the same cell. Complementary pure-judged and combat-to-semi classification variants (β_HS = +18.99 and +26.40) are documented in the analysis code and give bootstrap p in the 0.15-0.65 range. The two normalized specifications (vi)-(vii) test whether the raw-score interaction is a scale artifact: (vi) preserves the positive direction with cluster-robust p = 0.003 and bootstrap p = 0.158, essentially matching the raw-score bootstrap p = 0.155 and arguing against the result being solely a raw-unit artifact, although the magnitude depends on the normalization; (vii) attenuates to bootstrap p = 0.498, indicating that the interaction is weaker under the rank scale than under the raw-score or z-score scales. Bootstrap CIs are*

Cameron-Gelbach-Miller (2008) pivotal; n.c. = not computed (row iv runs through the `run_bootstrap_by_variant` *code path which does not emit a cluster-robust CI). Bootstrap uses Rademacher weights on prefecture clusters, B = 999, restricted null, seed 20260728 (shared across all bootstrap variants). Primary sample (rows i-iii, v-vii) retains all 41 sports (33 common to 2024 Saga and 2025 Shiga plus 8 edition-specific, e.g. clay-target shooting appended in 2024 and removed in 2025); row (iv) restricts to 19 sports (17 objective + 体操 + 馬術) for n = 3,359. R^2 omitted for row (iv) because the variant is computed through the bootstrap runner rather than the OLS runner.*

Table 7. Descriptive host-boost by Balmer et al. (2003) sport category (2024-2025 stacked cross-section; per-sport score)

Category	Mean non-host	Mean host	Cells (non-host)	Cells (host)	Host boost (points)
Objective (17 sports)	26.03	43.63	2,920	65	**+17.60**
Semi-subjective / team (13 sports)	20.85	47.03	2,199	48	**+26.18**
Subjective (11 sports)	19.33	57.24	1,721	38	**+37.91**

Monotonic increase in raw host boost across the three Balmer et al. (2003) sport categories, in exactly the direction the decomposition predicts. The subjective category's +37.91 raw score points is numerically larger than the objective category's +17.60; the raw-score ratio of 2.15 is descriptive and should not be interpreted as a proportional bonus without normalization for sport-specific total-score ceilings (see Limitations §Fifth). Figure 3 displays these three category-level host boosts with 95% confidence intervals from per-category `score ~ is_host` *OLS with prefecture-clustered SEs (47 clusters).*

Table 8. Cross-national context: host-scoring institutional arrangements

System	Host score bonus	Empirical host effect	Reference
Japan Kokutai (present study)	None (no institutional bonus in JSPO scoring)	Emperor's-cup host top-1 rate 76.0% (57 of 75) vs. chance 2.13%; +16.68 subjective-vs-reference interaction on a +16.60 combined objective / semi-subjective reference-category base per sport (Table 6 primary; 2024-2025 subsidiary; wild-cluster bootstrap p = 0.155)	Present study
Korea national comprehensive sports	20% score bonus (10% from 2001, raised to 20% in 2010)	Institutionally amplified	Kim (2025), Munhwa Ilbo[8]
China National Games	(institutional bonus not documented in accessible sources)	Qualitatively described (five characteristics: high efficacy, gradual increase, delayed effect, exclusivity, double-edgedness)	Lan & Yu (2012)[11]
Summer Olympics	None	Attenuates ~45% under population + GDP + country FE	Csurilla & Fertő (2023)[9]

*Among the four cases summarized here, Japan is the clearest example in which a large host-prefecture top-*k *concentration is observed in the absence of any institutional score bonus. Whether this concentration reflects a genuine causal home-advantage effect or systematic correlation between the host role and prefecture-level competitive strength cannot be identified from the observed pattern alone (see Limitations Primary conceptual limitation), and no socioeconomic-controls-augmented analysis is undertaken in the present study (ESRI 2008SNA prefectural GDP series limited to 2011-2022; see Limitations §First).*

Supplementary Materials

§S1. Mammen weights sensitivity check

As a robustness check for the wild-cluster bootstrap inference reported in Table 6, we re-estimated the primary (inclusive) specification bootstrap using Mammen (1993) two-point discrete asymmetric weights[16] in place of the Rademacher weights used for the headline bootstrap. The Mammen distribution takes the values $-(\sqrt{5}-1)/2 \approx -0.618$ with probability $(\sqrt{5}+1)/(2\sqrt{5}) \approx 0.724$ and $(\sqrt{5}+1)/2 \approx 1.618$ with probability $(\sqrt{5}-1)/(2\sqrt{5}) \approx 0.276$, giving $E[w] = 0$ and $E[w^2] = E[w^3] = 1$. This differs from Rademacher weights (which satisfy $E[w^3] = 0$) in preserving third-moment information — a distinction that can matter when the score residual distribution is skewed. Holding the seed, cluster structure, and restricted-null fit unchanged, the Mammen-weighted bootstrap p-value was **0.181** for the primary (inclusive) specification, compared with the Rademacher-weighted value of **0.155** reported in Table 6. Both weight families yield bootstrap p-values in the 0.15-0.19 range and neither reaches conventional significance under the two-treated-cluster correction, so the qualitative reading of Table 6 — that the point estimate is directionally consistent with the Balmer et al. (2003) decomposition but not statistically significant under the appropriate few-treated-clusters correction — is unchanged by the choice of weight distribution within the wild bootstrap family. The observed coefficients and t-statistics are unchanged by construction, since the wild-cluster bootstrap re-weights the restricted-null residuals rather than the observed fit.

§S2. Funahashi et al. (2016) Replication (2003-2011 Panel)

The v3 main analyses use the 75-non-cancelled-edition host-rank panel (n = 150) and focus on host top-*k* proportions rather than on the total-score OLS specification of Funahashi, Hibino, Ishiguro & Mano (2016).[6] For continuity with the Japanese-language predecessor literature, we retain the Funahashi replication in this supplement as a stand-alone pipeline validation, executed on the 2003-2011 legacy 47-prefecture-year panel (n = 423) that predates the v3 host-rank specification. This replication is implemented in `src/analysis_replication.py::run_replication_models()` and dumped in `results/analysis_replication.txt`; it is retained in the analysis repository for reproducibility but is not part of the v3 host-rank main analyses reported in the body of the paper.

Replication specification. We fit **Funahashi (2016)'s Table 4** base specification (referring to Table 4 in the Funahashi *et al.* 2016 paper, not to Table 4 of the present manuscript, which is the three-era χ^2 test) — prefecture-and-year fixed-effect OLS on the emperor's-cup total score with a single host-year dummy — omitting Funahashi's population, GDP, headquarters, transfer-athlete, and participant controls because the Cabinet Office ESRI 2008SNA prefectural GDP series does not extend before fiscal 2011. Standard errors are prefecture-clustered on 47 clusters.

Result. Our reproduction yields a host coefficient of **+1,575.45** total-score points (SE = 57.58; 47 prefecture clusters; $p < 10^{-40}$ under the safety cap applied elsewhere in this manuscript to hyper-precise exponents; $R^2 = 0.94$), compared with Funahashi's reported +1,674.65 — **within 5.9% of Funahashi's estimate** (|1,674.65 − 1,575.45| / 1,674.65 = 5.92%).

Sensitivity checks. A pooled OLS without fixed effects gives +1,733.29 (SE = 138.26); a +1-year extension to 2003-2012 (n = 470) gives +1,604.36 (SE = 60.08). Together, the three specifications bracket Funahashi's headline value of +1,674.65 from both above (pooled_no_fe: +1,733.29) and below (funahashi_base: +1,575.45; extended_2003_2012: +1,604.36). Standardized against each specification's own cluster-robust SE, the deviations from Funahashi's reference are 0.42 SE (pooled_no_fe), 1.17 SE (extended_2003_2012), and 1.72 SE (funahashi_base) — differences within the range attributable to the omitted socioeconomic controls (see Omitted-controls caveat below).

Extension through 2022 (legacy). As a further pipeline validation on the same 47-prefecture-year panel structure that Funahashi (2016) used, we extend the panel to 2012-2022 (n = 423) and fit a pooled ordered logit on rank (1-8, with "outside top 8" coded as rank 9) and a pooled binary logit on top-1 (championship) status. The pooled ordered logit yields a host coefficient of **−13.73** (SE = 2.03; two-sided normal-approximation $p \approx 1.4 \times 10^{-11}$, computed from $z = 13.73/2.03 \approx 6.76$ because the results dump prints `p=0.000000` at 6-decimal precision; negative meaning better rank), and the pooled binary logit for top-1 gives **+9.09** (SE = 2.13; $p = 1.9 \times 10^{-5}$), both on the 2012-2022 legacy panel (results archived at commit `fa07fd2` in the git history of the same GitHub repository, under `results/analysis_main.txt` generated by `run_all_models.py` invoking `src/analysis_main.py` in that commit). These pooled specifications sidestep the small-sample separation of the host indicator that renders Funahashi's original prefecture-and-year fixed-effect OLS degenerate on top-1 in the golden era (the fixed-effect binary logit for top-1 gives an inflated +1,877 with a numerically undefined SE — a classic separation blowup — reported in the archived `results/analysis_main.txt` at commit `fa07fd2` for transparency only). This legacy 2012-2022 panel and the pipeline that generated it are archived in the git history at commit `fa07fd2` and earlier of the same GitHub repository for reproducibility but are not part of the v3 host-rank main analyses reported in the body of the paper (see Limitations §Fourth for the parallel treatment of the Brant partial-proportional-odds diagnostic on the same legacy panel).

Omitted-controls caveat. The 5.9% gap relative to Funahashi's headline coefficient is well within what would be expected from omitting the socioeconomic controls. A formal all-controls-included replication would require assembly of pre-2011 prefectural socioeconomic covariates from a separate ESRI historical release, which was outside the scope of the present study. This omission does not affect the v3 host-rank main analyses (which do not use Funahashi's OLS specification), and this replication is retained here purely for pipeline validation against a published Japanese-language predecessor estimate.

R^2 caveat. Our reduced-control replication reports $R^2 = 0.94$ while Funahashi (2016) reports $R^2 = 0.86$ for their fully-controlled specification. Under standard OLS on an identical sample, dropping controls cannot *raise* R^2 (variance-decomposition arithmetic), so the observed direction implies at least one of: (a) the two runs are computed on non-identical samples (differences in how partial-participation prefecture-year cells and winter-only-sport zero cells are handled between our reproduction and Funahashi's original); (b) the two runs use different R^2 definitions (e.g., raw R^2 vs. adjusted R^2, or a within-transformation R^2 vs. an overall R^2); (c) our reproduction differs from Funahashi's fitted specification in ways beyond the omitted covariates (e.g., inclusion or exclusion of the winter-meet sports, treatment of the empress's-cup rows, or clustering choices that flow into the reported R^2 via a robust variance formula). We cannot fully adjudicate among (a)-(c) without direct access to Funahashi's raw sample and R^2 computation code, both of which are unavailable to us. We therefore report the R^2 gap as an open discrepancy rather than a mechanically explained artifact. The R^2 gap is not the load-bearing comparison for the replication — the host-coefficient agreement to within 5.9% of Funahashi's published +1,674.65 is — and does not affect the v3 main-body analyses, which do not rely on this replication's R^2 for any substantive claim.

Supplementary Figure S1. Funahashi (2016) replication host coefficient comparison across three specifications: (1) `funahashi_base` = 2003-2011 with prefecture and year fixed effects (n = 423; coefficient +1,575.45, cluster-robust SE = 57.58), (2) `pooled_no_fe` = 2003-2011 without fixed effects (n = 423; coefficient +1,733.29, cluster-robust SE = 138.26), and (3) `extended_2003_2012` = 2003-2012 with prefecture and year fixed effects (n = 470; coefficient +1,604.36, cluster-robust SE = 60.08). Error bars = 95% cluster-robust confidence intervals. The vertical dashed line indicates Funahashi (2016)'s published headline coefficient (+1,674.65). All three specifications bracket the published value: `pooled_no_fe` (+1,733.29) lies 0.42 SE above, and `funahashi_base` (+1,575.45) and `extended_2003_2012` (+1,604.36) lie 1.72 SE and 1.17 SE below, respectively (see §S2 Sensitivity checks for the standardized-deviation methodology and the Omitted-controls caveat for the interpretation of the 5.9% gap relative to the published headline value).

§S3. Sensitivity to 46-vs-47 State Null (Pre-Okinawa Reversion, 1972)

Motivation. The primary one-sample proportion test uses a null rate of $k/47$ (47 prefectures). Prior to Okinawa's 1972 reversion to Japan, only 46 prefectures competed in the Kokutai (editions 3-26, spanning 1948-1971 = 24 of the 30 early-era editions). The default $k/47$ null therefore slightly underestimates the correct chance baseline for these 24 pre-reversion editions, and hence slightly *overestimates* the excess-over-chance figure for the early era.

Method. As a conservative worst-case bound, we recompute the one-sample proportion test for the full 30-edition early era (editions 3-32, 1948-1977) under a 46-state null (rather than the actual mixed 46/47 structure), applied uniformly to all 30 editions. This yields an upper bound on how much the default 47-

state specification could be biasing the reported excess-over-chance figures for the early era. The implementation is in `src/analysis_main_v3.py::one_sample_proportion_test()` with the `n_states=46` parameter, and the dump is available in `results/analysis_main_v3.txt` section [2b].

Result. Across all 9 threshold × cup cells for the early era, the excess-over-chance under the 46-state null differs from the 47-state baseline by less than 0.005 (absolute). The full 9-cell table is reproduced verbatim from `results/analysis_main_v3.txt` [2b]:

threshold	cup	era	n	obs	obs_rate	null_rate	excess	p_greater	CI_low	CI_high
1	tennou	early	30	17	0.5667	0.0217	0.5449	$< 10^{-5}$	0.392	0.726
3	tennou	early	30	25	0.8333	0.0652	0.7681	$< 10^{-5}$	0.664	0.927
8	tennou	early	30	26	0.8667	0.1739	0.6928	$< 10^{-5}$	0.703	0.947
1	kougou	early	30	9	0.3000	0.0217	0.2783	$< 10^{-5}$	0.167	0.479
3	kougou	early	30	20	0.6667	0.0652	0.6014	$< 10^{-5}$	0.488	0.808
8	kougou	early	30	26	0.8667	0.1739	0.6928	$< 10^{-5}$	0.703	0.947
1	both	early	60	26	0.4333	0.0217	0.4116	$< 10^{-5}$	0.316	0.559
3	both	early	60	45	0.7500	0.0652	0.6848	$< 10^{-5}$	0.628	0.842
8	both	early	60	52	0.8667	0.1739	0.6928	$< 10^{-5}$	0.758	0.931

Comparison against 47-state baseline. For the corresponding early-era cells under the default *k*/47 null (dump section [2] of `results/analysis_main_v3.txt`), the excess figures are 0.5454 (top-1 tennou), 0.7695 (top-3 tennou), and 0.6965 (top-8 tennou) — differences of 0.0005, 0.0014, and 0.0037 respectively relative to the 46-state figures above. The default 47-state null therefore mildly overstates excess for the early era, but the impact is marginal (< 0.005 across all threshold-cup cells), and the qualitative reading of §Results — that host prefectures reach top-*k* at rates substantially above chance across all thresholds — is preserved unchanged. This limitation is referenced in Limitations §Third.

§S4. Era boundary sensitivity: 30-combination grid on the top-1 non-monotonic pattern

Motivation. The 1978 golden-era start and 2016 shock-era start boundaries used in Table 4 were selected with knowledge of the observed outcome data (Methods §Study design 3; Limitations §Third). As a boundary-sensitivity check we recompute the per-cup top-1 rate and the conditional Pearson χ^2 exact p for the three-era comparison on every combination of `golden_start ∈ {1975, 1976, 1977, 1978, 1979, 1980}` and `shock_start ∈ {2014, 2015, 2016, 2017, 2018}` — 6 × 5 = 30 boundary pairs × 2 cups = 60 fits. The dump is available at `results/analysis_main_v3.txt` [4c] and is implemented in `src/analysis_main_v3.py::era_boundary_sensitivity_grid`.

Result. Across all 30 boundary pairs on both cups, the three-era pattern satisfies `rate_golden > rate_early` AND `rate_golden > rate_shock` — the non-monotonic golden-era spike survives every considered boundary shift. Across the same 60 fits, the conditional Pearson χ^2 exact p is below 0.05 in every cell; the maximum observed exact p in the grid is well below 0.01 for both cups. Representative values from the corner combinations:

golden_start	shock_start	cup	rate_early	rate_golden	rate_shock	exact p
1975	2014	emperor's	0.519	0.974	0.556	2.94×10^{-5}
1975	2014	empress's	0.296	0.923	0.333	3.37×10^{-8}
1978	2016	emperor's	0.567	0.974	0.429	3.63×10^{-5}
1978	2016	empress's	0.300	0.921	0.429	5.38×10^{-8}
1980	2018	emperor's	0.567	0.925	0.600	6.65×10^{-4}
1980	2018	empress's	0.320	0.833	0.600	8.77×10^{-6}

The non-monotonic top-1 pattern is therefore not an artifact of the specific 1978 / 2016 boundary choice; it is preserved with statistical significance across the full ± 3-year neighborhood on both cups. The pattern would be undermined only by boundary combinations that leave the shock era with too few observations to identify (e.g., shock_start = 2020, which would leave n_shock = 3 non-cancelled editions) — such boundaries are outside the grid tested here because they would not permit a meaningful three-era comparison.

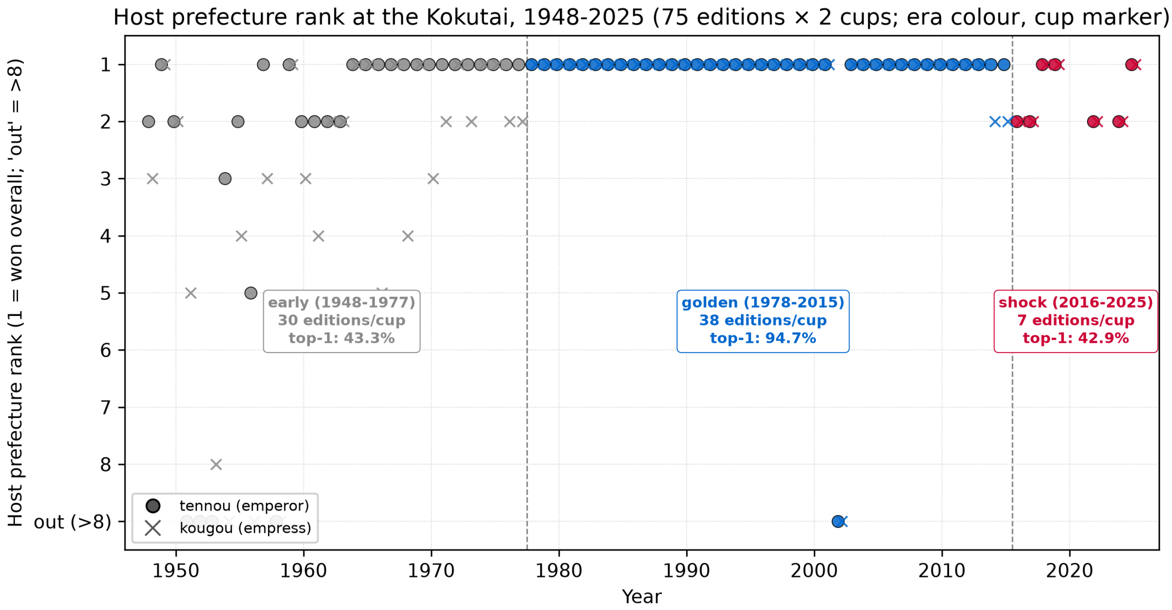


Figure 1. Host prefecture rank at the Kokutai, 1948-2025 (75 editions × 2 cups; era colour coding, cup marker). Each marker is one host-edition-cup observation (n = 150). Marker colour indicates era (early 1948-1977 grey; golden 1978-2015 blue; shock 2016-2025 red). Marker shape indicates cup (emperor's cup filled circle; empress's cup open x). Vertical dashed lines mark the era boundaries (1978 and 2016). The y-axis is inverted so that rank 1 sits at the top and "out (>8)" sits at the bottom; observations with `host_rank` outside the top 8 are placed at the "out" level (9 in the ordinal encoding). Ten `host_rank` observations sit at the "out" level, distributed as follows: 1951 Hiroshima (emperor's cup), 1952 Fukushima (both cups; a primary-host normalization of a multi-prefecture co-host), 1953 Ehime emperor's cup (empress's cup finished at rank 8), 1954 Hokkaido (empress's cup), 1956 Hyōgo (empress's cup), 1958 Toyama (both cups), and 2002 Kōchi (both cups). The 2002 Kōchi observation is the sole "out" event outside the early era; it corresponds to the pre-2005 furusato-athlete-rule shock and is the only golden-era edition on which the primary host finished outside the top 8 on either cup. The per-era host top-1 rate is annotated on each era's coloured box (early 43.3%; golden 94.7%; shock 42.9%; both cups pooled), making the non-monotonic time pattern immediately visible.

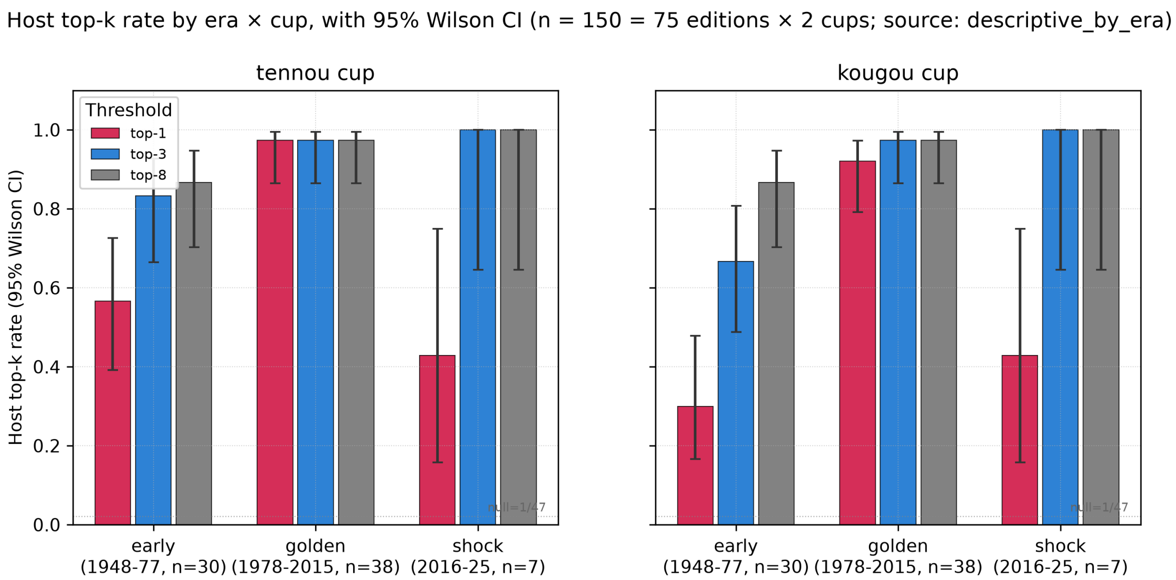


Figure 2. Host top-k rate by era × cup × threshold, with 95% Wilson two-sided confidence intervals. Left panel: emperor's cup. Right panel: empress's cup. Within each panel, three bars per era correspond to the top-1 (red), top-3 (blue), and top-8 (grey) thresholds. Error bars are Wilson two-sided 95% confidence intervals from `scipy.stats.binomtest(...).proportion_ci(method='wilson')` applied to the observed n_success / n on each cell. The horizontal dotted reference line at y = 1/47 marks the top-1 chance baseline (2.13%); the top-3 and top-8 chance baselines (6.38% and 17.02%) are also below every observed rate on every cell but are not drawn as reference lines to preserve visual clarity. Source: `descriptive_by_era` in Table 1 combined with `one_sample_proportion_test` for the CI extraction. n = 150 (both cups pooled across 75 editions).

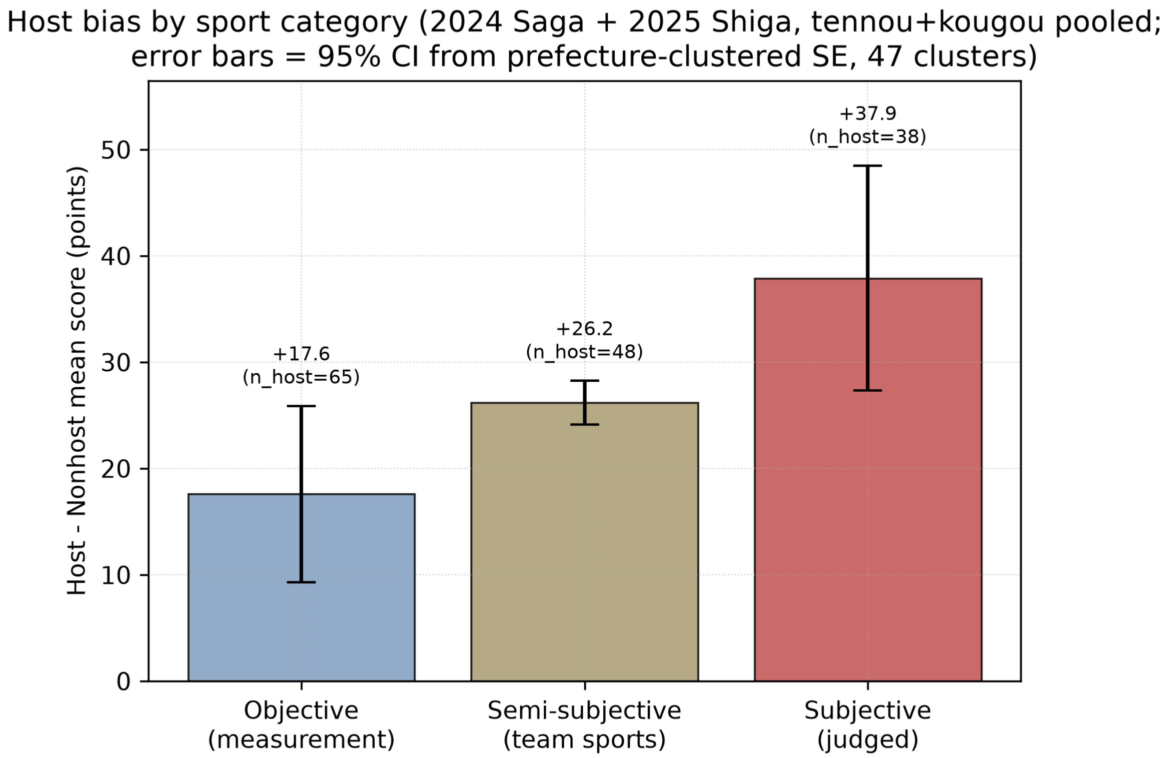
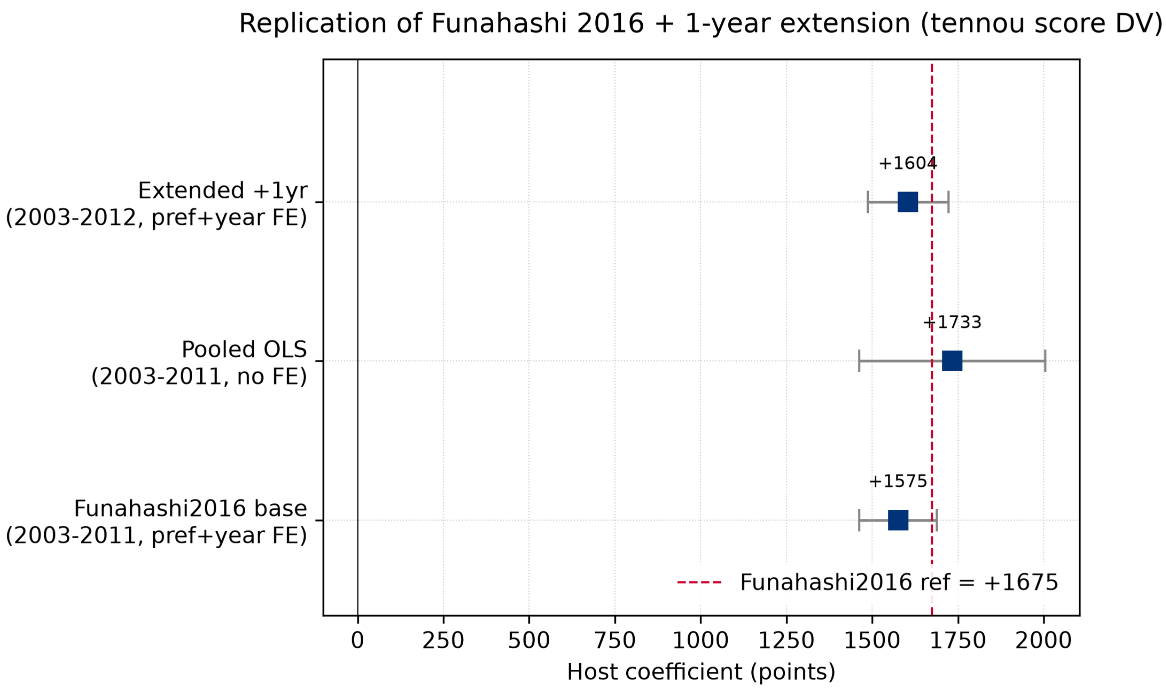


Figure 3. Host-versus-non-host per-sport score gap by Balmer et al. (2003) sport category, in the 2024-2025 stacked cross-section (subsidiary analysis, n = 6,991). Bars show the mean host boost in points; error bars show **descriptive naive cluster-robust 95% confidence intervals** (1.96 × prefecture-clustered SE from per-category `score ~ is_host` OLS with SE clustered on `pref_code`, 47 clusters), reported here as a per-category descriptive band only. Because only 2 of 47 prefecture clusters are treated (Saga in 2024 and Shiga in 2025), the same few-treated-clusters correction described in Methods §Subsidiary applies to these per-category confidence intervals as well; they are known to be anti-conservative under the two-treated-cluster structure and are not corrected by a wild-cluster bootstrap in this figure — the joint host × subjective interaction test that carries the primary inferential statement is reported in Table 6 (rows i-ii, cluster-robust p = 0.031 vs. wild-cluster bootstrap p = 0.155). Objective sports (17 sports, 65 host cells): +17.60 points. Semi-subjective / team sports (13 sports, 48 host cells): +26.18 points. Subjective sports (11 sports, 38 host cells): +37.91 points. The monotonic ordering matches the Balmer et al. (2003) directional prediction; the +37.91 / +17.60 raw-score ratio of 2.15 is descriptive and does not adjust for sport-specific total-score ceilings or participant-count differences (see Limitations §Fifth).

Supplementary Figures



Supplementary Figure S1. Funahashi (2016) replication host coefficient comparison across three specifications: (1) `funahashi_base` = 2003-2011 with prefecture and year fixed effects (n = 423; coefficient +1,575.45, cluster-robust SE = 57.58), (2) `pooled_no_fe` = 2003-2011 without fixed effects (n = 423; coefficient +1,733.29, cluster-robust SE = 138.26), and (3) `extended_2003_2012` = 2003-2012 with prefecture and year fixed effects (n = 470; coefficient +1,604.36, cluster-robust SE = 60.08). Error bars = 95% cluster-robust confidence intervals. The vertical dashed line indicates Funahashi (2016)'s published headline coefficient (+1,674.65). All three specifications bracket the published value: `pooled_no_fe` (+1,733.29) lies 0.42 SE above, and `funahashi_base` (+1,575.45) and `extended_2003_2012` (+1,604.36) lie 1.72 SE and 1.17 SE below, respectively (see §S2 Sensitivity checks for the standardized-deviation methodology and the Omitted-controls caveat for the interpretation of the 5.9% gap relative to the published headline value).